



US 20250287357A1

(19) **United States**

(12) **Patent Application Publication**
Zhang et al.

(10) **Pub. No.: US 2025/0287357 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **TIME DOMAIN CHANNEL PROPERTY REPORTING**

(52) **U.S. Cl.**
CPC *H04W 72/0446* (2013.01); *H04B 7/0626* (2013.01); *H04W 24/10* (2013.01)

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(57) **ABSTRACT**

(21) Appl. No.: **18/862,470**

(22) PCT Filed: **Aug. 12, 2022**

(86) PCT No.: **PCT/CN2022/112265**

§ 371 (c)(1),

(2) Date: **Nov. 1, 2024**

Publication Classification

(51) **Int. Cl.**

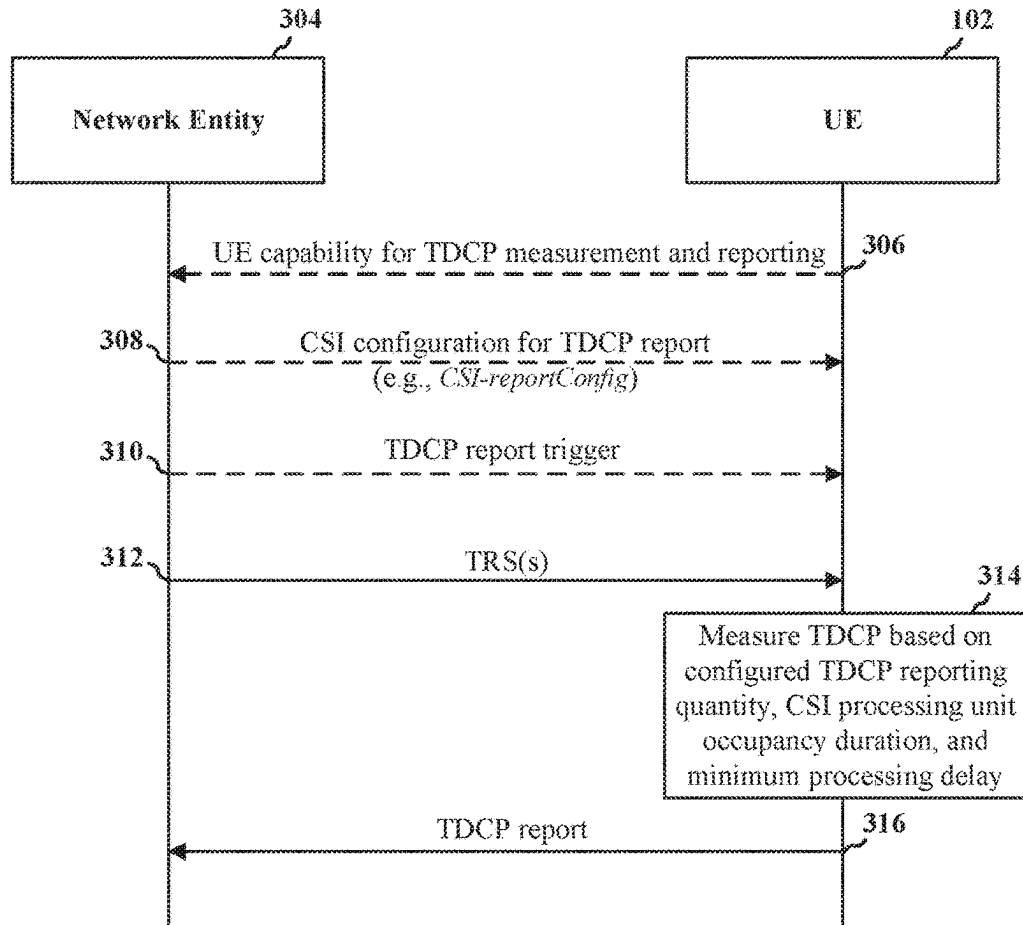
H04W 72/0446 (2023.01)

H04B 7/06 (2006.01)

H04W 24/10 (2009.01)

This disclosure provides systems, devices, apparatus, and methods, including computer programs encoded on storage media, for TDCP measurement and reporting techniques. A UE (102) receives (312) a TRS from a network entity (304) through a channel between the network entity (304) and the UE (102). The UE (102) measures, based on the received (312) TRS, a TDCP of the channel between the network entity (304) and the UE (102). A measurement (314) of the TDCP of the channel is associated with at least one of: a configured reporting quantity for a TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. The UE (102) transmits (316) the TDCP report to the network entity (304) based on the measurement (314).

300



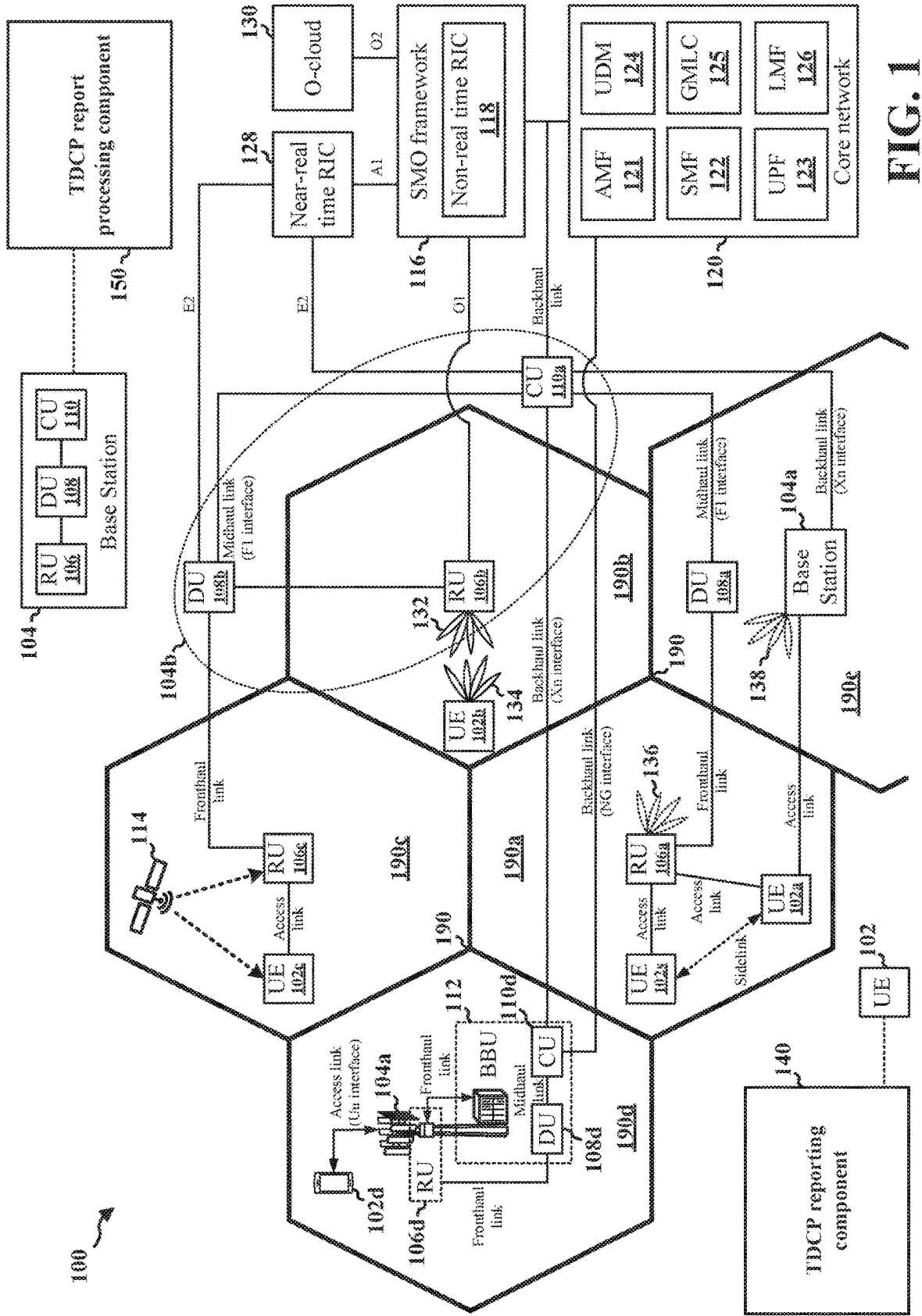


FIG. 1

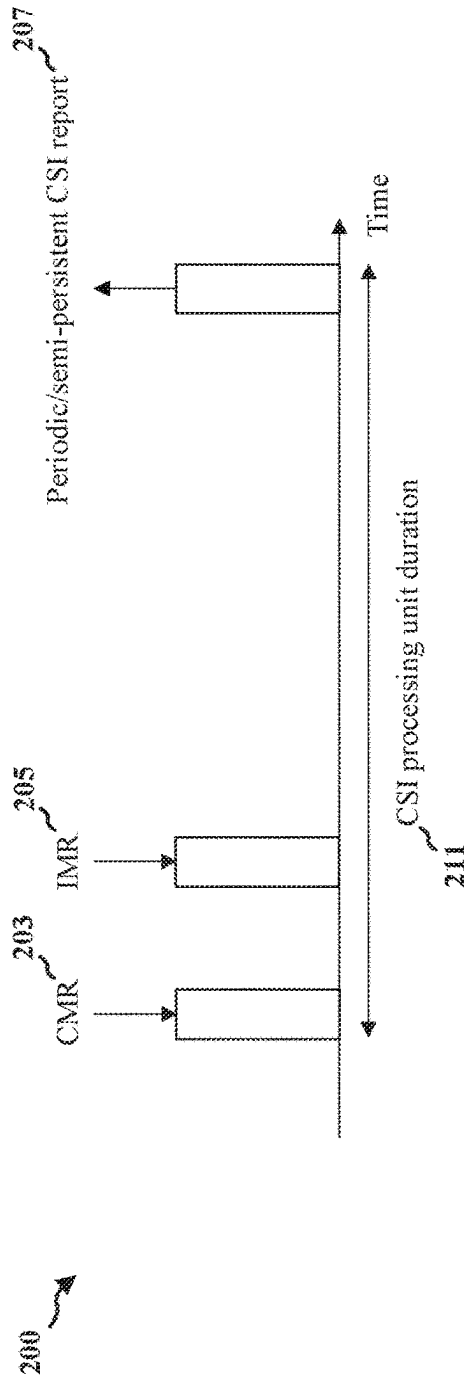


FIG. 2A

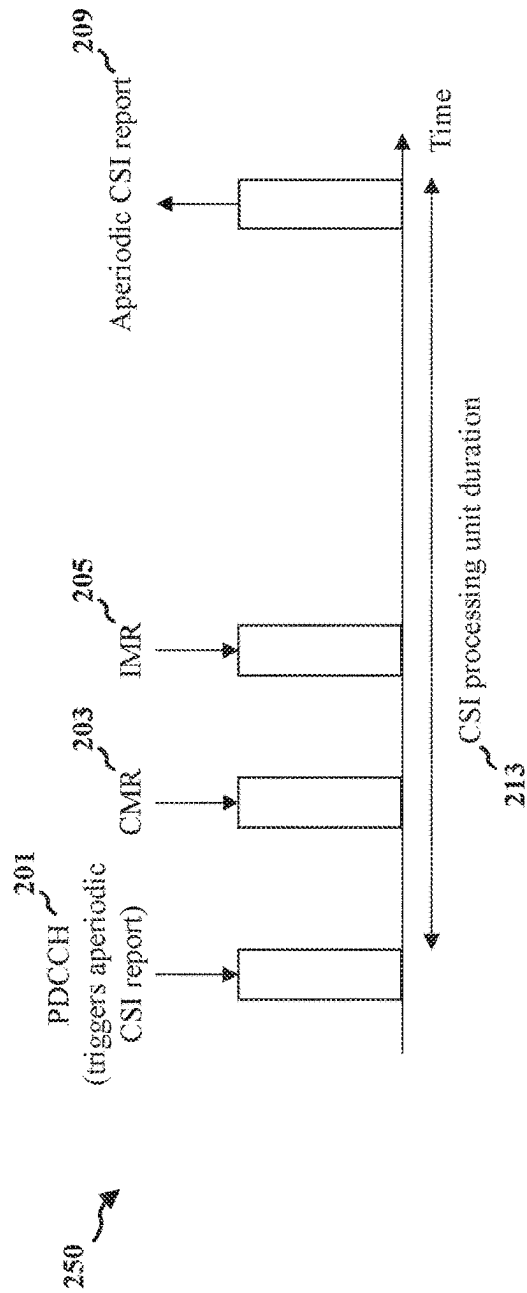


FIG. 2B

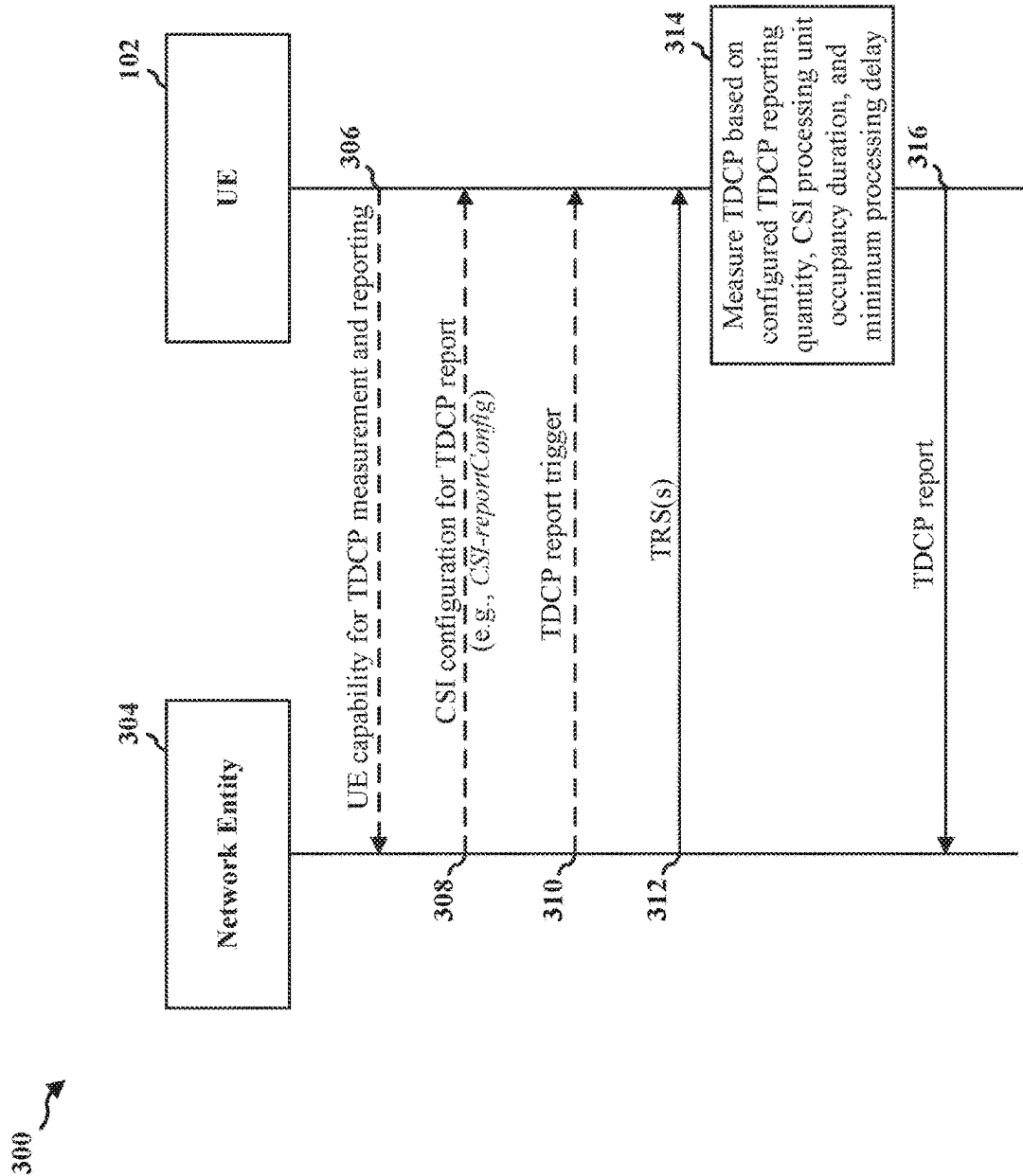


FIG. 3

410 ↗

CSI-RS resource set index 1, if reported
CSI-RS resource set index 2, if reported
...
CSI-RS resource set index V, if reported
TDCP for CSI-RS resource set index 1
TDCP for CSI-RS resource set index 2, if reported
...
TDCP for CSI-RS resource set index V, if reported

FIG. 4A

420 ↗

CSI-RS resource set index 1, if reported
CSI-RS resource set index 2, if reported
...
CSI-RS resource set index V, if reported
First TDCP for CSI-RS resource set index 1
First TDCP for CSI-RS resource set index 2, if reported
...
First TDCP for CSI-RS resource set index V, if reported
Second TDCP for CSI-RS resource set index 1
Second TDCP for CSI-RS resource set index 2, if reported
...
Second TDCP for CSI-RS resource set index V, if reported

FIG. 4B

430 ↗

CSI-RS resource set index 1, if reported
CSI-RS resource set index 2, if reported
...
CSI-RS resource set index V, if reported
L1-RSRP for CSI-RS resource set index 1
Differential L1-RSRP for CSI-RS resource set index 2, if reported
...
Differential L1-RSRP for CSI-RS resource set index V, if reported
TDCP for CSI-RS resource set index 1
TDCP for CSI-RS resource set index 2, if reported
...
TDCP for CSI-RS resource set index V, if reported

FIG. 4C

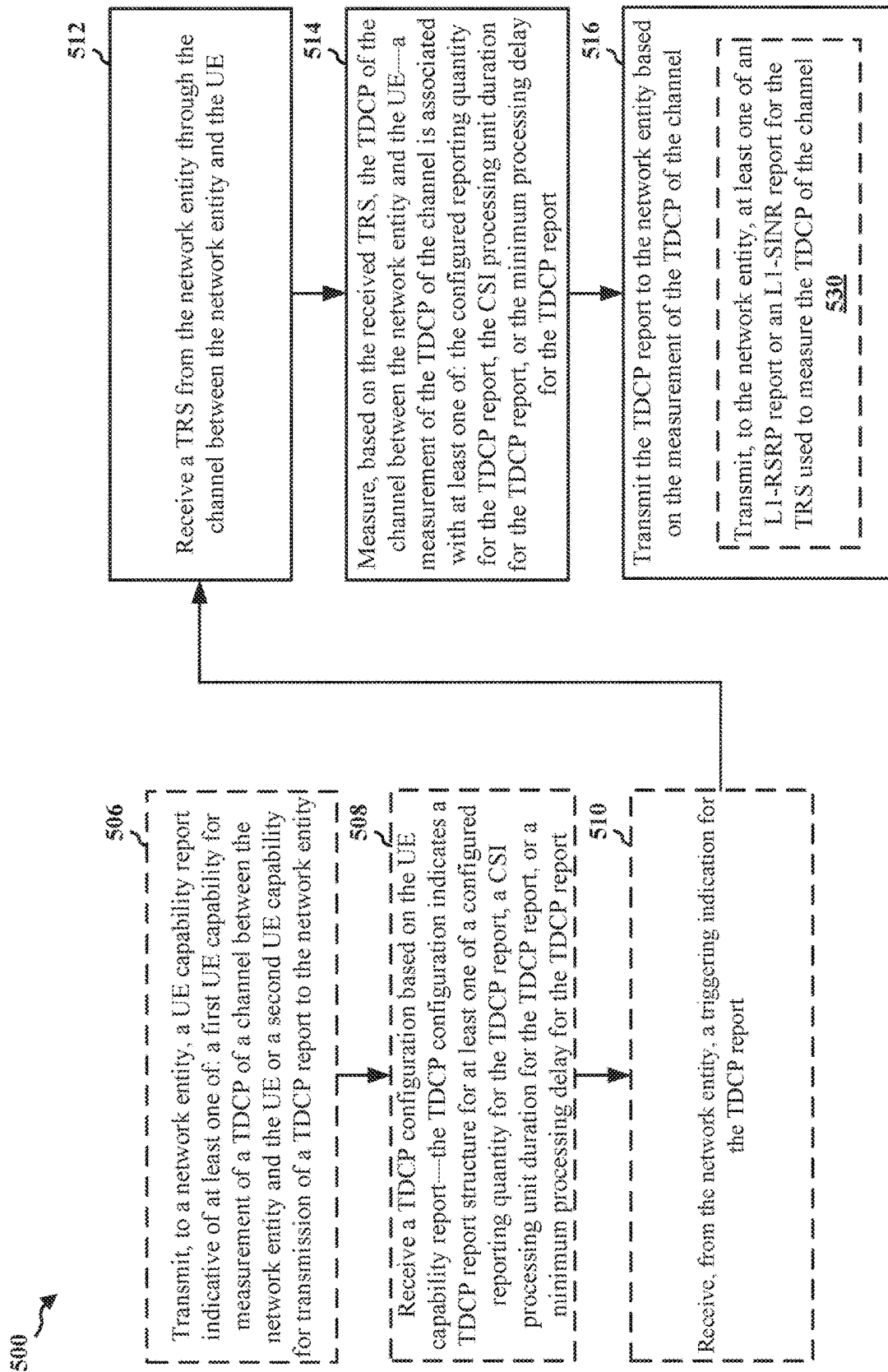


FIG. 5

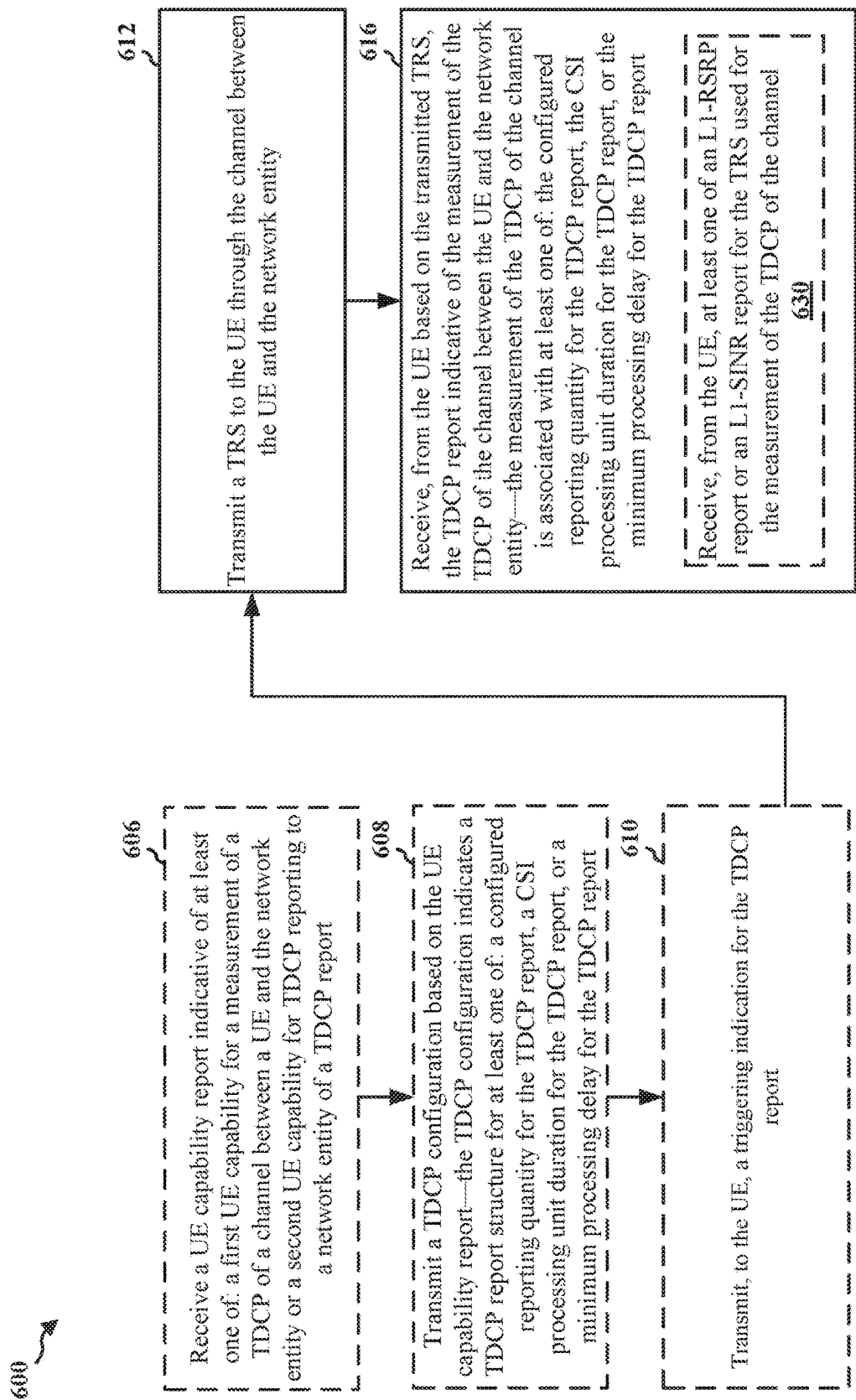
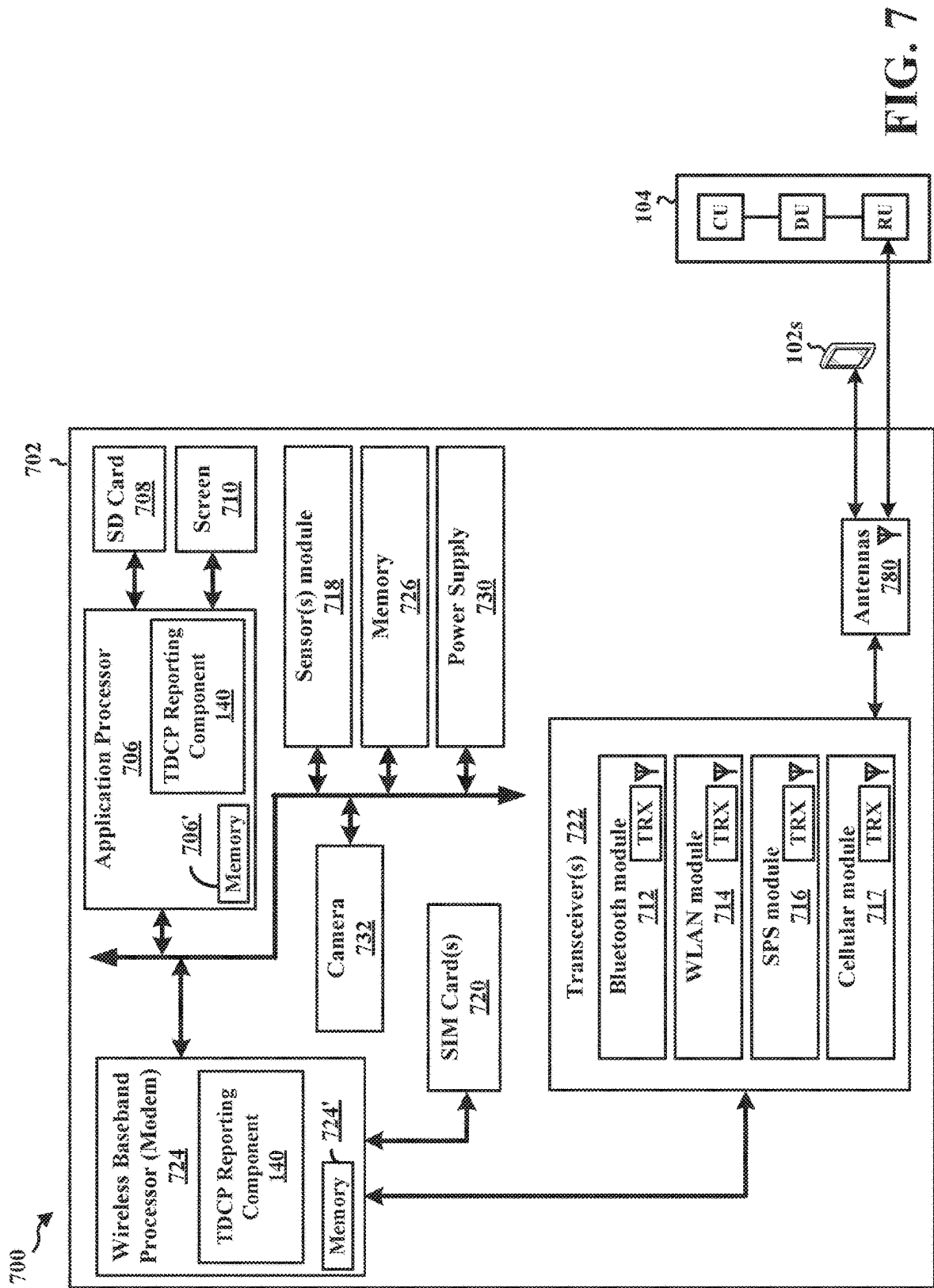


FIG. 6



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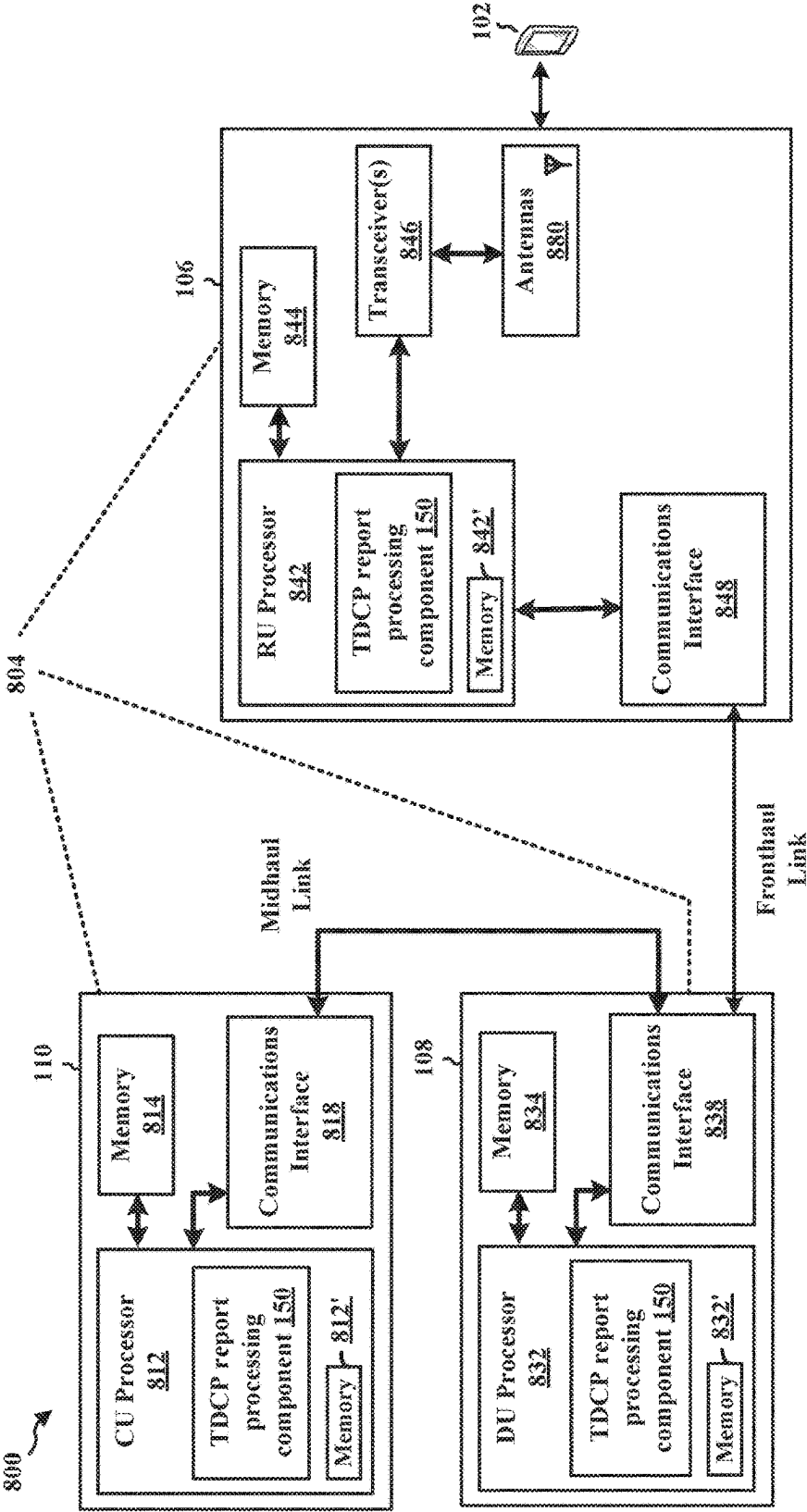


FIG. 8

TIME DOMAIN CHANNEL PROPERTY REPORTING

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a 35 U.S.C. § 371 application of International Application No. PCT/CN2022/112265, filed Aug. 12, 2022, which is incorporated herein by reference in their entirety and for all purposes.

TECHNICAL FIELD

[0002] The present disclosure relates generally to wireless communication, and more particularly, to time domain channel property (TDCP) measurement and reporting techniques.

BACKGROUND

[0003] The Third Generation Partnership Project (3GPP) specifies a radio interface referred to as fifth generation (5G) new radio (NR) (5G NR). An architecture for a 5G NR wireless communication system can include a 5G core (5GC) network, a 5G radio access network (5G-RAN), a user equipment (UE), etc. The 5G NR architecture might provide increased data rates, decreased latency, and/or increased capacity compared to other types of wireless communication systems. The documents “CSI enhancement for high/medium UE velocities and CJT” (Nokia et al.), 3GPP Draft—R1-2204540, 3rd Generation Partnership Project, vol. RAN WG1, no. e-Meeting; 20220509-20220520 and “On Rel-18 CSI enhancements”, 3GPP Draft; R1-2203443, 3rd Generation Partnership Project (CATT), vol. RAN WG1, no. e-Meeting; 20220509-20220520 29 Apr. 2022 (2022 Apr. 29), XP052152975 relate to this radio interface.

[0004] Wireless communication systems, in general, may be configured to provide various telecommunication services (e.g., telephony, video, data, messaging, broadcasts, etc.) based on multiple-access technologies, such as orthogonal frequency division multiple access (OFDMA) technologies, that support communication with multiple UEs. As mobile broadband technologies evolve, improvements in mobile broadband have been useful to continue the progression of such technologies. For example, a user equipment (UE) that is moving relative to a base station might experience a phase shift in a channel path between the UE and the base station that causes a quality of the channel path to degrade. WO 2022/162507 proposes apparatuses, methods, and systems for configuring tracking reference signal resources. One method includes receiving an indication of a HST SFN transmission from at least one TRP in a network. The method includes configuring the UE with two TRS resources. The method includes receiving a downlink scheduling grant that includes a TCI codepoint indicating two TCI states corresponding to the two TRS resources. The method includes configuring the UE to report at least one Doppler indicator value to the network based on received TRSs. The method includes identifying the at least one Doppler indicator value. The method includes reporting the at least one Doppler indicator value to the network. The method includes receiving (514) at least one DMRS port corresponding to each layer of a PDSCH, PDCCH, or a combination thereof. US2019/165847 proposes a method of reporting, by a UE, CSI in a wireless communication system, the method comprising: receiving, from a base station, a RRC signaling that

comprises a plurality of reporting settings, wherein each reporting setting comprises a corresponding list of first values representing time offsets for transmitting a CSI report, forming a plurality of lists of first values; receiving, from the base station, DCI triggering the CSI report, wherein the DCI comprises an index value related to a time at which to transmit the CSI report on a PUSCH; determining, based on the DCI, a plurality of list entries; determining a second value that is largest among the plurality of list entries; and transmitting, to the base station, the CSI report on the PUSCH based on the second value. WO 2017/171928 proposes mechanisms for a legacy UE traveling at a high speed (e.g., in a high speed train) to network-assisted mobility status indication and tracking for a UE traveling at a high speed (e.g., in a high speed train). It also provides for monitoring an uplink (UL) transmission signal received from a User Equipment (UE) in the SFN at a plurality of remote radio heads (RRHs). A mobility status of the UE according to the UL transmission signal may be determined. Channel propagation conditions and mobility status may be sent in a downlink transmission signal for transmission to the UE.

BRIEF SUMMARY

[0005] The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects. This summary neither identifies key or critical elements of all aspects nor delineates the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

[0006] A phase shift in a channel path between a user equipment (UE) and a network entity, such as a base station, may be based on a moving speed of the UE toward or away from the network entity, a carrier frequency, an angle of arrival (AoA) for the channel path, etc. Although each channel path of a multi-path channel might be associated with a same speed of the UE and a same carrier frequency, different AoAs for different channels paths of a multi-path channel might create different phase shifts for each channel path.

[0007] The UE may receive a tracking reference signal (TRS) from the network entity to perform time and frequency offset tracking, such as Doppler shift estimations and/or Doppler spread estimations associated with downlink decoding. However, after the UE receives and measures the TRS from the network entity, the UE might not be configured to report time domain channel properties (TDCPs), to the network entity, based on the TRS. That is, the UE may internally utilize the measured TDCPs (e.g., Doppler shift estimations and/or Doppler spread estimations) without indicating the measured TDCPs to the network entity. Meanwhile, the network entity relies on channel state information (CSI) measured and reported by the UE to perform procedures such as downlink scheduling. For example, in a multiple-input multiple-output (MIMO) system, the network entity may indicate a digital precoder to the UE based on a CSI report received from the UE. However, if the UE is moving at an increased rate of speed relative to the network entity (e.g., large Doppler shift), the CSI report received from the UE may not be sufficiently reliable for the network entity to efficiently schedule downlink transmissions for the

UE, which may result in reduced transmission efficiency between the network entity and the UE.

[0008] Aspects of the present disclosure address the above-noted and other deficiencies by configuring the UE to transmit a TDCP report to the network entity for assisting the network entity with downlink scheduling procedures for the UE.

[0009] According to some aspects, the UE receives a TRS from a network entity through a channel between the network entity and the UE. The UE measures, based on the received TRS, a TDCP of the channel between the network entity and the UE. A measurement of the TDCP of the channel is associated with at least one of: a configured reporting quantity for a TDCP report or a CSI processing unit duration for the TDCP report. The UE transmits the TDCP report to the network entity based on the measuring of the TDCP of the channel based on the received TRS.

[0010] According to some aspects, the network entity transmits a TRS to a UE through a channel between the UE and the network entity. The network entity receives, from the UE based on the transmitted TRS, a TDCP report indicative of a measurement of the TDCP of the channel between the UE and the network entity. The measurement of the TDCP of the channel is associated with at least one of: a configured reporting quantity for the TDCP report or a CSI processing unit duration for the TDCP report.

[0011] To the accomplishment of the foregoing and related ends, the one or more aspects correspond to the features hereinafter described and particularly pointed out in the claims. The one or more aspects may be implemented through any of an apparatus, a method, a means for performing the method, and/or a non-transitory computer-readable medium. The following description and the drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 illustrates a diagram of a wireless communications system including a plurality of network entities in communication over a plurality of cells.

[0013] FIG. 2A illustrates a first timing diagram for a first channel state information (CSI) processing unit duration associated with periodic/semi-persistent CSI reporting.

[0014] FIG. 2B illustrates a second timing diagram for a second CSI processing unit duration associated with aperiodic CSI reporting.

[0015] FIG. 3 is a signaling diagram illustrating communications between a user equipment (UE) and a network entity for a time domain channel property (TDCP) measurement and reporting procedure.

[0016] FIGS. 4A-4C illustrate a plurality of tables that include example TDCP report information transmitted from the UE to the network entity.

[0017] FIG. 5 is a flowchart of a method of TDCP reporting at a UE.

[0018] FIG. 6 is a flowchart of a method of TDCP reporting at a network entity.

[0019] FIG. 7 is a diagram illustrating an example of a hardware implementation for an example UE apparatus.

[0020] FIG. 8 is a diagram illustrating an example of a hardware implementation for one or more example network entities.

DETAILED DESCRIPTION

[0021] FIG. 1 illustrates a diagram 100 of a wireless communications system associated with a plurality of cells 190. The wireless communications system includes user equipments (UEs) 102 and base stations 104, where some base stations 104a include an aggregated base station architecture and other base stations 104b include a disaggregated base station architecture. The aggregated base station architecture includes a radio unit (RU) 106, a distributed unit (DU) 108, and a centralized unit (CU) 110 that are configured to utilize a radio protocol stack that is physically or logically integrated within a single radio access network (RAN) node. A disaggregated base station architecture utilizes a protocol stack that is physically or logically distributed among two or more units (e.g., RUs 106, DUs 108, CUs 110). For example, a CU 110 is implemented within a RAN node, and one or more DUs 108 may be co-located with the CU 110, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs 108 may be implemented to communicate with one or more RUs 106. Each of the RU 106, the DU 108 and the CU 110 can be implemented as virtual units, such as a virtual radio unit (VRU), a virtual distributed unit (VDU), or a virtual central unit (VCU).

[0022] Operations of the base stations 104 and/or network designs may be based on aggregation characteristics of base station functionality. For example, disaggregated base station architectures are utilized in an integrated access backhaul (IAB) network, an open-radio access network (O-RAN) network, or a virtualized radio access network (vRAN) which may also be referred to a cloud radio access network (C-RAN). Disaggregation may include distributing functionality across the two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network designs. The various units of the disaggregated base station architecture, or the disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit. For example, the CU 110a communicates with the DUs 108a-108b via respective midhaul links based on F1 interfaces. The DUs 108a-108b may respectively communicate with the RU 106a and the RUs 106b-106c via respective fronthaul links. The RUs 106a-106c may communicate with respective UEs 102a-102c and 102s via one or more radio frequency (RF) access links based on a Uu interface. In examples, multiple RUs 106 and/or base stations 104 may simultaneously serve the UEs 102, such as the UE 102a of the cell 190a that the access links for the RU 106a of the cell 190a and the base station 104a of the cell 190e simultaneously serve.

[0023] One or more CUs 110, such as the CU 110a or the CU 110d, may communicate directly with a core network 120 via a backhaul link. For example, the CU 110d communicates with the core network 120 over a backhaul link based on a next generation (NG) interface. The one or more CUs 110 may also communicate indirectly with the core network 120 through one or more disaggregated base station units, such as a near-real time RAN intelligent controller (RIC) 128 via an E2 link and a service management and orchestration (SMO) framework 116, which may be associated with a non-real time RIC 118. The near-real time RIC 128 might communicate with the SMO framework 116 and/or the non-real time RIC 118 via an A1 link. The SMO framework 116 and/or the non-real time RIC 118 might also

communicate with an open cloud (O-cloud) **130** via an O2 link. The one or more CUs **110** may further communicate with each other over a backhaul link based on an Xn interface. For example, the CU **110d** of the base station **104a** communicates with the CU **110a** of the base station **104b** over the backhaul link based on the Xn interface. Similarly, the base station **104a** of the cell **190e** may communicate with the CU **110a** of the base station **104b** over a backhaul link based on the Xn interface.

[0024] The RUs **106**, the DUs **108**, and the CUs **110**, as well as the near-real time RIC **128**, the non-real time RIC **118**, and/or the SMO framework **116**, may include (or may be coupled to) one or more interfaces configured to transmit or receive information/signals via a wired or wireless transmission medium. A base station **104** or any of the one or more disaggregated base station units can be configured to communicate with one or more other base stations **104** or one or more other disaggregated base station units via the wired or wireless transmission medium. In examples, a processor, a memory, and/or a controller associated with executable instructions for the interfaces can be configured to provide communication between the base stations **104** and/or the one or more disaggregated base station units via the wired or wireless transmission medium. For example, a wired interface can be configured to transmit or receive the information/signals over a wired transmission medium, such as for the fronthaul link between the RU **106d** and the baseband unit (BBU) **112** of the cell **190d** or, more specifically, the fronthaul link between the RU **106d** and DU **108d**. The BBU **112** includes the DU **108d** and a CU **110d**, which may also have a wired interface configured between the DU **108d** and the CU **110d** to transmit or receive the information/signals between the DU **108d** and the CU **110d** based on a midhaul link. In further examples, a wireless interface, which may include a receiver, a transmitter, or a transceiver (such as an RF transceiver), can be configured to transmit or receive the information/signals via the wireless transmission medium, such as for information communicated between the RU **106a** of the cell **190a** and the base station **104a** of the cell **190e** via cross-cell communication beams of the RU **106a** and the base station **104a**.

[0025] One or more higher layer control functions, such as function related to radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), and the like, may be hosted at the CU **110**. Each control function may be associated with an interface for communicating signals based on one or more other control functions hosted at the CU **110**. User plane functionality such as central unit-user plane (CU-UP) functionality, control plane functionality such as central unit-control plane (CU-CP) functionality, or a combination thereof may be implemented based on the CU **110**. For example, the CU **110** can include a logical split between one or more CU-UP procedures and/or one or more CU-CP procedures. The CU-UP functionality may be based on bidirectional communication with the CU-CP functionality via an interface, such as an E1 interface (not shown), when implemented in an O-RAN configuration.

[0026] The CU **110** may communicate with the DU **108** for network control and signaling. The DU **108** is a logical unit of the base station **104** configured to perform one or more base station functionalities. For example, the DU **108** can control the operations of one or more RUs **106**. One or more of a radio link control (RLC) layer, a medium access

control (MAC) layer, or one or more higher physical (PHY) layers, such as forward error correction (FEC) modules for encoding/decoding, scrambling, modulation/demodulation, or the like can be hosted at the DU **108**. The DU **108** may host such functionalities based on a functional split of the DU **108**. The DU **108** may similarly host one or more lower PHY layers, where each lower layer or module may be implemented based on an interface for communications with other layers and modules hosted at the DU **108**, or based on control functions hosted at the CU **110**.

[0027] The RUs **106** may be configured to implement lower layer functionality. For example, the RU **106** is controlled by the DU **108** and may correspond to a logical node that hosts RF processing functions, or lower layer PHY functionality, such as execution of fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, etc. The functionality of the RUs **106** may be based on the functional split, such as a functional split of lower layers.

[0028] The RUs **106** may transmit or receive over-the-air (OTA) communication with one or more UEs **102**. For example, the RU **106b** of the cell **190b** communicates with the UE **102b** of the cell **190b** via a first set of communication beams **132** of the RU **106b** and a second set of communication beams **134** of the UE **102b**, which may correspond to inter-cell communication beams or cross-cell communication beams. Both real-time and non-real-time features of control plane and user plane communications of the RUs **106** can be controlled by associated DUs **108**. Accordingly, the DUs **108** and the CUs **110** can be utilized in a cloud-based RAN architecture, such as a vRAN architecture, whereas the SMO framework **116** can be utilized to support non-virtualized and virtualized RAN network elements. For non-virtualized network elements, the SMO framework **116** may support deployment of dedicated physical resources for RAN coverage, where the dedicated physical resources may be managed through an operations and maintenance interface, such as an O1 interface. For virtualized network elements, the SMO Framework **116** may interact with a cloud computing platform, such as the O-cloud **130** via the O2 link (e.g., cloud computing platform interface), to manage the network elements. Virtualized network elements can include, but are not limited to, RUs **106**, DUs **108**, CUs **110**, near-real time RICs **128**, etc.

[0029] The SMO framework **116** may be configured to utilize an O1 link to communicate directly with one or more RUs **106**. The non-real time RIC **118** of the SMO framework **116** may also be configured to support functionalities of the SMO framework **116**. For example, the non-real time RIC **118** implements logical functionality that enables control of non-real time RAN features and resources, features/applications of the near-real time RIC **128**, and/or artificial intelligence/machine learning (AI/ML) procedures. The non-real time RIC **118** may communicate with (or be coupled to) the near-real time RIC **128**, such as through the A1 interface. The near-real time RIC **128** may implement logical functionality that enables control of near-real time RAN features and resources based on data collection and interactions over an E2 interface, such as the E2 interfaces between the near-real time RIC **128** and the CU **110a** and the DU **108b**.

[0030] The non-real time RIC **118** may receive parameters or other information from external servers to generate AI/ML models for deployment in the near-real time RIC

128. For example, the non-real time RIC **118** receives the parameters or other information from the O-cloud **130** via the O2 link for deployment of the AI/ML models to the real-time RIC **128** via the A1 link. The near-real time RIC **128** may utilize the parameters and/or other information received from the non-real time RIC **118** or the SMO framework **116** via the A1 link to perform near-real time functionalities. The near-real time RIC **128** and the non-real time RIC **115** may be configured to adjust a performance of the RAN. For example, the non-real time RIC **116** monitors patterns and long-term trends to increase the performance of the RAN. The non-real time RIC **116** may also deploy AI/ML models for implementing corrective actions through the SMO framework **116**, such as initiating a reconfiguration of the O1 link or indicating management procedures for the A1 link.

[0031] Any combination of the RU **106**, the DU **108**, and the CU **110**, or reference thereto individually, may correspond to a base station **104**. Hence, the base station **104** may include at least one of the RU **106**, the DU **108**, or the CU **110**. The base stations **104** provide the UEs **102** with access to the core network **120**. That is, the base stations **104** might relay communications between the UEs **102** and the core network **120**. The base stations **104** may be associated with macrocells for high-power cellular base stations and/or small cells for low-power cellular base stations. For example, the cell **190e** corresponds to a macrocell, whereas the cells **190a-190d** may correspond to small cells. Small cells include femtocells, picocells, microcells, etc. A cell structure that includes at least one macrocell and at least one small cell may be referred to as a “heterogeneous network.”

[0032] Transmissions from a UE **102** to a base station **104**/RU **106** are referred to uplink (UL) transmissions, whereas transmissions from the base station **104**/RU **106** to the UE **102** are referred to as downlink (DL) transmissions. Uplink transmissions may also be referred to as reverse link transmissions and downlink transmissions may also be referred to as forward link transmissions. For example, the RU **106d** utilizes antennas of the base station **104a** of cell **190d** to transmit a downlink/forward link communication to the UE **102d** or receive an uplink/reverse link communication from the UE **102d** based on the Uu interface associated with the access link between the UE **102d** and the base station **104a**/RU **106d**.

[0033] Communication links between the UEs **102** and the base stations **104**/RUs **106** may be based on multiple-input multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be associated with one or more carriers. The UEs **102** and the base stations **104**/RUs **106** may utilize a spectrum bandwidth of Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) per carrier allocated in a carrier aggregation of up to a total of Yx MHz, where x component carriers (CCs) are used for communication in each of the uplink and downlink directions. The carriers may or may not be adjacent to each other along a frequency spectrum. In examples, uplink and downlink carriers may be allocated in an asymmetric manner, more or fewer carriers may be allocated to either the uplink or the downlink. A primary component carrier and one or more secondary component carriers may be included in the component carriers. The primary component carrier may be associated with a primary cell (PCell) and a secondary component carrier may be associated with as a secondary cell (SCell).

[0034] Some UEs **102**, such as the UEs **102a** and **102s**, may perform device-to-device (D2D) communications over sidelink. For example, a sidelink communication/D2D link utilizes a spectrum for a wireless wide area network (WWAN) associated with uplink and downlink communications. The sidelink communication/D2D link may also use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and/or a physical sidelink control channel (PSCCH), to communicate information between UEs **102a** and **102s**. Such sidelink/D2D communication may be performed through various wireless communications systems, such as wireless fidelity (Wi-Fi) systems, Bluetooth systems, Long Term Evolution (LTE) systems, New Radio (NR) systems, etc.

[0035] The electromagnetic spectrum is often subdivided into different classes, bands, channels, etc., based on different frequencies/wavelengths associated with the electromagnetic spectrum. Fifth-generation (5G) NR is generally associated with two operating bands referred to as frequency range 1 (FR1) and frequency range 2 (FR2). FR1 ranges from 410 MHz-7.125 GHz and FR2 ranges from 24.25 GHz-52.6 GHz. Although a portion of FR1 is actually greater than 6 GHz, FR1 is often referred to as the “sub-6 GHz” band. In contrast, FR2 is often referred to as the “millimeter wave” (mmW) band. FR2 is different from, but a near subset of, the “extremely high frequency” (EHF) band, which ranges from 30 GHz-300 GHz and is sometimes also referred to as a “millimeter wave” band. Frequencies between FR1 and FR2 are often referred to as “mid-band” frequencies. The operating band for the mid-band frequencies may be referred to as frequency range 3 (FR3), which ranges 7.125 GHz-24.25 GHz. Frequency bands within FR3 may include characteristics of FR1 and/or FR2. Hence, features of FR1 and/or FR2 may be extended into the mid-band frequencies. Higher operating bands have been identified to extend 5G NR communications above 52.6 GHz associated with the upper limit of FR2. Three of these higher operating bands include FR2-2, which ranges from 52.6 GHz-71 GHz, FR4, which ranges from 71 GHz-114.25 GHz, and FR5, which ranges from 114.25 GHz-300 GHz. The upper limit of FR5 corresponds to the upper limit of the EHF band. Thus, unless otherwise specifically stated herein, the term “sub-6 GHz” may refer to frequencies that are less than 6 GHz, within FR1, or may include the mid-band frequencies. Further, unless otherwise specifically stated herein, the term “millimeter wave”, or mmW, refers to frequencies that may include the mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.

[0036] The UEs **102** and the base stations **104**/RUs **106** may each include a plurality of antennas. The plurality of antennas may correspond to antenna elements, antenna panels, and/or antenna arrays that may facilitate beamforming operations. For example, the RU **106b** transmits a downlink beamformed signal based on a first set of beams **132** to the UE **102b** in one or more transmit directions of the RU **106b**. The UE **102b** may receive the downlink beamformed signal based on a second set of beams **134** from the RU **106b** in one or more receive directions of the UE **102b**. In a further example, the UE **102b** may also transmit an uplink beamformed signal to the RU **106b** based on the second set of beams **134** in one or more transmit directions

of the UE 102b. The RU 106b may receive the uplink beamformed signal from the UE 102b in one or more receive directions of the RU 106b. The UE 102b may perform beam training to determine the best receive and transmit directions for the beam formed signals. The transmit and receive directions for the UEs 102 and the base stations 104/RUs 106 might or might not be the same. In further examples, beamformed signals may be communicated between a first base station 104a and a second base station 104b. For instance, the RU 106a of cell 190a may transmit a beamformed signal based on an RU beam set 136 to the base station 104a of cell 190e in one or more transmit directions of the RU 106a. The base station 104a of the cell 190e may receive the beamformed signal from the RU 106a based on a base station beam set 138 in one or more receive directions of the base station 104a. Similarly, the base station 104a of the cell 190e may transmit a beamformed signal to the RU 106a based on the base station beam set 138 in one or more transmit directions of the base station 104a. The RU 106a may receive the beamformed signal from the base station 104a of the cell 190e based on the RU beam set 136 in one or more receive directions of the RU 106a.

[0037] The base station 104 may include and/or be referred to as a next generation evolved Node B (ng-eNB), a generation NB (gNB), an evolved NB (eNB), an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmit reception point (TRP), a network node, a network entity, network equipment, or other related terminology. The base station 104 or an entity at the base station 104 can be implemented as an IAB node, a relay node, a sidelink node, an aggregated (monolithic) base station with an RU 106 and a BBU that includes a DU 108 and a CU 110, or as a disaggregated base station 104b including one or more of the RU 106, the DU 108, and/or the CU 110. A set of aggregated or disaggregated base stations 104a-104b may be referred to as a next generation-radio access network (NG-RAN).

[0038] The core network 120 may include an Access and Mobility Management Function (AMF) 121, a Session Management Function (SMF) 122, a User Plane Function (UPF) 123, a Unified Data Management (UDM) 124, a Gateway Mobile Location Center (GMLC) 125, and/or a Location Management Function (LMF) 126. The core network 120 may also include one or more location servers, which may include the GMLC 125 and the LMF 126, as well as other functional entities. For example, the one or more location servers include one or more location/positioning servers, which may include the GMLC 125 and the LMF 126 in addition to one or more of a position determination entity (PDE), a serving mobile location center (SMLC), a mobile positioning center (MPC), or the like.

[0039] The AMF 121 is the control node that processes the signaling between the UEs 102 and the core network 120. The AMF 121 supports registration management, connection management, mobility management, and other functions. The SMF 122 supports session management and other functions. The UPF 123 supports packet routing, packet forwarding, and other functions. The UDM 124 supports the generation of authentication and key agreement (AKA) credentials, user identification handling, access authorization, and subscription management. The GMLC 125 provides an interface for clients/applications (e.g., emergency services) for accessing UE positioning information. The

LMF 126 receives measurements and assistance information from the NG-RAN and the UEs 102 via the AMF 121 to compute the position of the UEs 102. The NG-RAN may utilize one or more positioning methods in order to determine the position of the UEs 102. Positioning the UEs 102 may involve signal measurements, a position estimate, and an optional velocity computation based on the measurements. The signal measurements may be made by the UEs 102 and/or the serving base stations 104/RUs 106.

[0040] Communicated signals may also be based on one or more of a satellite positioning system (SPS) 114, such as signals measured for positioning. In an example, the SPS 114 of the cell 190c may be in communication with one or more UEs 102, such as the UE 102c, and one or more base stations 104/RUs 106, such as the RU 106c. The SPS 114 may correspond to one or more of a Global Navigation Satellite System (GNSS), a global position system (GPS), a non-terrestrial network (NTN), or other satellite position/location system. The SPS 114 may be associated with LTE signals, NR signals (e.g., based on round trip time (RTT) and/or multi-RTT), wireless local area network (WLAN) signals, a terrestrial beacon system (TBS), sensor-based information, NR enhanced cell ID (NR E-CID) techniques, downlink angle-of-departure (DL-AoD), downlink time difference of arrival (DL-TDOA), uplink time difference of arrival (UL-TDOA), uplink angle-of-arrival (UL-AoA), and/or other systems, signals, or sensors.

[0041] The UEs 102 may be configured as a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a GPS, a multimedia device, a video device, a digital audio player (e.g., moving picture experts group (MPEG) audio layer-3 (MP3) player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an utility meter, a gas pump, appliances, a healthcare device, a sensor/actuator, a display, or any other device of similar functionality. Some of the UEs 102 may be referred to as Internet of Things (IoT) devices, such as parking meters, gas pumps, appliances, vehicles, healthcare equipment, etc. The UE 102 may also be referred to as a station (STA), a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a mobile client, a client, or other similar terminology. The term UE may also apply to a roadside unit (RSU), which may communicate with other RSU UEs, non-RSU UEs, a base station 104, and/or an entity at a base station 104, such as an RU 106.

[0042] Still referring to FIG. 1, in certain aspects, the UE 102 may include a time domain channel property (TDCP) reporting component 140 configured to receive a tracking reference signal (TRS) from a network entity through a channel between the network entity and a UE 102. The TDCP reporting component 140 is further configured to measure, based on the received TRS, a TDCP of the channel between the network entity and the UE 102. A measurement of the TDCP of the channel is associated with at least one of: a configured reporting quantity for a TDCP report, a channel state information (CSI) processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. The TDCP reporting component 140 is further con-

figured to transmit the TDCP report to the network entity based on the measuring the TDCP of the channel.

[0043] In certain aspects, the base station **104** or a network entity of the base station **104** may include a TDCP report processing component **150** configured to transmit a TRS to a UE through a channel between the UE and the network entity. The TDCP report processing component **150** is further configured to receive, from the UE based on the transmitted TRS, a TDCP report indicative of a measurement of the TDCP of the channel between the UE and the network entity. The measurement of the TDCP of the channel is associated with at least one of: a configured reporting quantity for the TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. Although the following description may be focused on 5G NR, the concepts described herein may be applicable to other similar areas, such as 5G-Advanced and future versions, LTE, LTE-advanced (LTE-A), and other wireless technologies.

[0044] FIG. 2A illustrates a first timing diagram **200** for a first CSI processing unit duration **211** associated with periodic/semi-persistent CSI reporting. FIG. 2B illustrates a second timing diagram **250** for a second CSI processing unit duration **213** associated with aperiodic CSI reporting. A UE **102** that is moving towards or away from a base station **104**, or a network entity of a base station **104**, may experience a phase shift (e.g., Doppler shift) in a channel path between the UE **102** and the base station **104** based on a Doppler effect associated with the movement of the UE **102**. The phase shift/Doppler shift may be based on a speed of the UE **102** relative to the base station **104**, a carrier frequency, an angle of arrival (AoA) for the channel path, etc. Although each channel path of a multi-path channel might be associated with a same speed of the UE **102** and a same carrier frequency, different AoAs for different channels paths of the multi-path channel might result in different Doppler shifts for each channel path.

[0045] The phase shift for a channel path k is calculated based on:

$$\gamma_k = \exp\left(j2\pi \frac{r_k \times v}{\lambda} t\right)$$

$$r_k = [\sin\theta_k \cos\varphi_k \quad \sin\theta_k \sin\varphi_k \quad \cos\theta_k]$$

$$v = v_0 [\sin\theta_v \cos\varphi_v \quad \sin\theta_v \sin\varphi_v \quad \cos\theta_v]^T$$

where λ indicates a waveform length, which is based on the carrier frequency, t indicates a time, θ_k indicates a zenith angle of arrival (ZAoA) for the channel path k , φ_k indicates the azimuth angle of arrival (AAoA) for the channel path k , v_0 indicates a moving speed of the UE **102**, θ_v indicates a first moving direction (e.g., vertical direction), and φ_v indicates a second moving direction (e.g., horizontal direction). The base station **104** may transmit a channel state information-reference signal (CSI-RS) to the UE **102** to perform time and frequency offset tracking for the movement of the UE **102**. The time and frequency offset tracking may be associated with Doppler shift estimations and/or Doppler spread estimations. The CSI-RS for the time and frequency offset tracking may be referred to as a TRS.

[0046] The TRS might correspond to a CSI-RS resource set indicated in an RRC message parameter, such as trs-Info. The base station **104** might configure a periodic TRS based

on RRC messaging (e.g., non-zero power(NZP)-CSI-RS-ResourceSet) using a higher layer parameter, such as the trs-Info. In further examples, the base station **104** may transmit an aperiodic TRS, indicated to the UE via downlink control information (DCI) signaling. The base station **104** may transmit the TRS in 2 symbols of a slot or in 4 symbols allocated to two consecutive slots, where the 2 CSI-RS resources or 4 CSI-RS resources are configured (e.g., in the NZP-CSI-RS-ResourceSet) using the higher layer parameter, such as the trs-Info. The UE **102** may estimate the Doppler shift and/or the Doppler spread for downlink decoding using the received TRS.

[0047] The aperiodic TRS may be quasi-co-located (QCLed) with the periodic TRS, and the base station **104** might provide a QCL indication via DCI. Further, the base station **104** might configure a QCL type and/or a source reference signal for the QCL signaling. QCL types for the source reference signal might be configured based on a higher layer parameter (e.g., the QCL type may correspond to a qcl-Type in a QCL-Info parameter). A first QCL type, typeA, includes values for a Doppler shift, a Doppler spread, an average delay, and a delay spread. A second QCL type, typeB, includes values for the Doppler shift and/or the Doppler spread. A third QCL type, typeC, includes values for the Doppler shift and/or the average delay. A fourth QCL type, typeD, includes values for a spatial receive (RX) parameter.

[0048] Communication links between the UEs **102** and the base stations **104** may be based on multiple-input multiple-output (MIMO) antenna technology. The base station **104** might select a digital precoder for the UE **102** based on a CSI report received from the UE **102**. The base station **104** may configure the CSI report using an information element (IE) of an RRC message (e.g., CSI-reportConfig), such that the UE **102** may use a channel measurement resource (CMR) **203** for measuring the downlink channel of a CSI-RS transmitted between the base station **104** and the UE **102**. Further, the base station **104** may configure an interference measurement resource (IMR) **205** for the UE **102** to measure interference. The UE can subsequently indicate the measured interference on the IMR in the CSI report to the base station **104** (e.g., based on the CSI-reportConfig parameter). The CMR **203** and the IMR **205** enable the UE **102** to measure CSI-RSs and subsequently report CSI to the base station **104**. The CSI report may include a rank indicator (RI), a precoder matrix indicator (PMI), a channel quality indicator (CQI), and/or a layer indicator (LI). The base station **104** may use the RI and the PMI to select the digital precoder for transmissions to the UE **102**. The CQI indicates a signal-to-interference-plus-noise ratio (SINR) for determining a modulation and coding scheme (MCS). The LI is indicative of a strongest layer, which may be used for multi-user (MU)-MIMO pairing with a low rank transmission and precoder selection for a phase-tracking reference signal (PT-RS).

[0049] The base station **104** might configure a time-domain CSI reporting technique to the UE **102**. For example, the base station **104** might configure the UE **102** for a periodic/semi-persistent CSI report **207** or an aperiodic CSI report **209** (e.g., using the CSI-reportConfig parameter). The base station **104** activates or deactivates the periodic/semi-persistent CSI report **207** using a medium access control-control element (MAC-CE) indication to the UE **102**. In further examples, the base station **104** might trigger

the aperiodic CSI report 209 by transmitting a physical downlink control channel (PDCCH) 201 instruction to the UE 102. The UE 102 may report periodic CSI on a physical uplink control channel (PUCCH) resource (e.g., configured based on the CSI-reportConfig parameter). The UE 102 may report semi-persistent CSI on the PUCCH resource or on a physical uplink shared channel (PUSCH) resource configured based on the DCI received from the base station 104. The UE 102 may also report aperiodic CSI on the PUSCH resource configured based on the DCI received from the base station 104.

[0050] The UE 102 can be configured with multiple CSI-reportConfig IEs for multiple CSI measurements and reports. Thus, a plurality of CSI processing units may be used for parallel processing of received CSI-RS to create a plurality of CSI measurements and reports. The UE 102 might transmit UE capability information to the base station 104 indicating a number of CSI processing units that the UE 102 supports. If the base station 104 requests CSI reports that require more parallel CSI processing at the UE 102 than the number of CSI processing units available at the UE (i.e., indicated in the UE capability information), the UE 102 can report non-current information for low priority CSI reports. In some examples, the UE 102 may determine a priority of a particular CSI report based on predefined protocols. For example, a CSI processing unit occupancy rule for the periodic/semi-persistent CSI report 207 and the aperiodic CSI report 209 may be based on the predefined protocols.

[0051] The first CSI processing unit duration 211 for the periodic/semi-persistent CSI report 207 corresponds to a CSI processing unit that's occupancy begins at a first symbol of earliest resources for the CMR 203 or the IMR 205 used for measurements by the UE 102. For example, the UE 102 may perform one or more measurements of a CSI-RS, CSI-interference measurement (CSI-IM), a synchronization signal block (SSB), etc. The first CSI processing unit duration 211 for the periodic/semi-persistent CSI report 207 continues through last resources for the CMR 203 and the IMR 205 used for the measurement of the UE 102 and ends at a last symbol of a PUSCH/PUCCH used by the UE 102 for transmitting the periodic/semi-persistent CSI report 207 to the base station 104.

[0052] The second CSI processing unit duration 213 for the aperiodic CSI report 209 corresponds to a CSI processing unit that's occupancy begins at a first symbol after receiving the PDCCH 201 that triggers the aperiodic CSI report 209. The second CSI processing unit duration 213 for the aperiodic CSI report 209 continues through last resources for the CMR 203 and the IMR 205 used for the measurement of the UE 102 and ends at a last symbol of a PUSCH used by the UE 102 for transmitting the aperiodic CSI report 209 to the base station 104. If the PDCCH 201 that triggers the aperiodic CSI report 209 corresponds to two PDCCH candidates from two respective search space sets, the PDCCH candidate that ends later in time is used for determining the second CSI processing unit duration 213 for the aperiodic CSI report 209.

[0053] If the PDCCH 201 triggers an initial semi-persistent CSI report on the PUSCH, the CSI processing unit duration for the initial semi-persistent CSI report should not be the same as the first CSI processing unit duration 211 for the periodic/semi-persistent CSI report 207. Instead, the CSI processing unit duration for the initial semi-persistent CSI report may correspond to the second CSI processing unit

duration 213 for the aperiodic CSI report 209. That is, the CSI processing unit duration for the initial semi-persistent CSI report transmitted on the PUSCH, after the PDCCH 201, begins at the first symbol after the PDCCH 201 and ends at the last symbol of the PUSCH that carries the initial semi-persistent CSI report. If the PDCCH 201 that triggers the initial semi-persistent CSI report corresponds to two PDCCH candidates from two respective search space sets, the PDCCH candidate that ends later in time is used for determining the CSI processing unit duration for the initial semi-persistent CSI report.

[0054] CSI reporting by the UE 102 might be based on a minimum processing delay time. For example, scheduling for the periodic/semi-persistent CSI report 207 or the aperiodic CSI report 209 might include minimum processing delays of Z and Z' . Values of Z and Z' for different types of CSI reports might be based on one or more predefined protocols. If a scheduling offset does not indicate the minimum processing delays of Z and Z' , the UE 102 can report the minimum processing delay associated with the non-current CSI or disregard a triggering DCI, if no other signals (e.g. data or hybrid automatic repeat request (HARQ)-acknowledgment (ACK) (HARQ-ACK)) are to be transmitted on the PUSCH triggered by the DCI (e.g., PDCCH 201). If a CSI request field of the DCI is used to trigger one or more CSI reports from the UE 102 on the PUSCH, the UE 102 may transmit the one or more CSI reports to the base station 104, if the first uplink symbol associated with the one or more CSI reports serves as a timing advance that starts no earlier than symbol Z_{ref} . For instance, the UE 102 may transmit an n -th triggered CSI report to the base station 104, if the first uplink symbol associated with the n -th CSI report starts no earlier than symbol $Z'_{ref}(n)$.

[0055] Some CSI feedback techniques might not include UE reporting of one or more TDCPs to the base station 104. Instead, the UE 102 may internally utilize the one or more TDCPs, such as Doppler shift and/or Doppler spread measured from the TRS, for CSI reporting procedures, but does not report the one or more TDCPs to the base station 104. A lack of information regarding TDCPs measured at the UE 102 may cause the base station 104 to rely on the CSI measured and reported to the base station 104 for performing further scheduling. If the UE 102 is moving at an increased rate of speed relative to the base station 104, an increased Doppler shift may occur, which might cause the CSI report received from the UE 102 to be inaccurate. Hence, the base station 104 might not be able to select a suitable precoder for the downlink transmission, which may reduce transmission efficiencies.

[0056] Accordingly, UE 102 transmission of a TDCP report to the base station 104 assists the base station 104 with the scheduling of the downlink transmissions for the UE 102. The base station 104 may determine based on the TDCP report whether the channel is experiencing significant change, so that the base station 104 may further determine whether to request additional CSI feedback or a sounding reference signal (SRS) for antenna beam control. If the TDCP report indicates a substantial change in channel conditions, the base station 104 may trigger the SRS for antenna switching to measure the downlink precoder based on uplink and downlink channel reciprocity. However, SRS techniques might increase system overhead based on the resources for the SRS being dedicated resources for the UE 102. If the TDCP report does not indicate a significant

channel change, the base station **104** may trigger the CSI report based on the PDCCH **201**. However, PDCCH triggering techniques may be associated with increased latency as shown by comparing FIG. 2B with FIG. 2A.

[0057] The base station **104** may generate a channel prediction or a channel Eigen vector prediction based on the reported TDCP received from the UE **102**, such that the base station **104** may further generate a second prediction for the precoder based on a past precoder matrix and the reported TDCP. The base station **104** transmits a triggering indication to the UE **102** to cause the UE **102** to transmit the TDCP report to the base station **104**. The UE **102** may determine the CSI processing unit occupancy rule as well as the minimum processing delay for the TDCP report after reception of the triggering indication. If the base station **104** configures a plurality of TRSs for TDCP measurements at the UE **102**, the UE **102** may select/determine the TDCPs to include in the TDCP report. FIGS. 2A-2B illustrate CSI reporting that does not include TDCP information. As will be discussed with respect to FIG. 3, the UE **102** may transmit a TDCP report including TDCP information to the network, so that the network can make scheduling determinations for channels associated with changing channel conditions.

[0058] FIG. 3 is a signaling diagram **300** illustrating communications between a UE **102** and a network entity **304** for a TDCP measurement and reporting procedure. The network entity **304** may correspond to the base station **104** or an entity at the base station **104**, such as the RU **106**, the DU **108**, the CU **110**, etc. The UE **102** may report **306** a UE capability to the network entity **304** for TDCP measurement and reporting. For example, the UE capability might correspond to a first UE capability for TDCP measurement and a second UE capability for TDCP reporting. The UE **102** might transmit **306** in UE capability information an indication of a first maximum number of TRSs that may be received **312** per CSI-reportConfig for the TDCP report, a second maximum number of TRSs that may be received **312** across the CSI-reportConfig for the TDCP report, and/or a third maximum number of CSI-reportConfig that may be received **312** for the TDCP report. The maximum numbers may be counted per component carrier (CC), per frequency band, per band combination, and/or per UE. The UE **102** may also transmit **306** an indication of parameters that are supported by the UE **102** for the TDCP measurement (e.g., a number of CSI-RS resources per TRS, a minimum bandwidth for the TRS, etc.).

[0059] The network entity **304** transmits **308** a CSI configuration to the UE **102** for the TDCP report based on the UE capability information received **306** from the UE **102**. For example, the CSI configuration transmitted **308** to the UE **102** can include a CMR configuration for the TDCP measurement. The network entity **304** might configure N TRS(s) based on RRC parameters (not shown; occurs after or in association with **308**). In examples, the RRC parameters may also be associated with an RRC reconfiguration message from the network entity **304** to the UE **102**, or a system information block (SIB). The SIB may correspond to an existing type of SIB (e.g., SIB1) or a different type of SIB (e.g., SIB J, where J is an integer above 21) transmitted by the network entity **304**. In further examples, the N TRS(s) may be configured for the TDCP measurement in the CSI-reportConfig. Thus, N NZP-CSI-RS-ResourceSet can be configured for resourcesForChannelMeasurement.

[0060] In certain examples, the network entity **304** configures N TRSs for the TDCP measurement based on the RRC message, but instructs the UE via MAC-CE or DCI to select M TRSs for performing the TDCP measurement, where $0 < M \leq N$. The network entity **304** may configure the N TRSs for the TDCP measurement based on the CSI-report-Config. Thus, N NZP-CSI-RS-ResourceSet can be configured for resourcesForChannelMeasurement, such that the network entity **304** may use the MAC-CE to indicate the M TRSs to be measured **314** at the UE **102**.

[0061] The MAC-CE may indicate a serving cell index, which may correspond to a serving cell index for the CSI-reportConfig for the TDCP measurement. In further examples, the MAC-CE may indicate a bandwidth part (BWP) index, which may correspond to a BWP index for the CSI-reportConfig for the TDCP measurement. In still further examples, the MAC-CE may indicate a CSI-reportConfigId, which may correspond to an identifier (ID) for a corresponding CSI-reportConfig for the TDCP measurement. In still further examples, the MAC-CE may indicate a bitmap for the TRS to be measured **314** at the UE **102**, which may correspond to an activation/deactivation status of the TDCP measurement for each TRS configured via the CSI-report-Config for the TDCP measurement. When using a triggering DCI, the network entity **304** transmits **310** a trigger to the UE **102** for the TDCP report. The network entity **304** may indicate different values in a CSI request field to indicate different TRSs that will be transmitted **312** to the UE **102** for the TDCP measurement. For instance, the network entity **304** may configure the different TRSs for the TDCP measurement for different CSI-AssociatedReportConfigInfo.

[0062] The network entity **304** determines the TRS transmitted **312** to the UE **102** for the TDCP measurement based on a transmission configuration indicator (TCI) state for the PDCCH/PDSCH associated with the TRS. The network entity **304** may indicate, in the TCI state, the TRS for QCL indication. The UE **102** measures **314** the TDCP based on reception **312** of the TRS from the network entity **304**. For example, the TRS transmitted **312** to the UE **102** might reflect channel distortions and multi-path between the UE **102** and the network entity **304**, such that the UE **102** may determine various channel properties based on the distortions. The UE **102** may use the TRS for QCL types other than typeD (associated with the spatial Rx parameter) to measure **314** the TDCP and transmit **316** the TDCP report.

[0063] The UE **102** might use the TRS for QCL typeD indication to perform the TDCP measurement and reporting when the TCI state for the PDCCH/PDSCH is dynamically changing. If multiple TCI states are indicated for the PDCCH/PDSCH, UE **102** may use the TRS in one of the TCI states for the TDCP measurement, such as the TRS in the TCI state with a lowest TCI state ID value. In other examples, the UE **102** may use all the TRSs in the TCI states for the TDCP measurement. If the UE **102** measures more than 1 TRS for generating the TDCP report, the UE **102** may measure **314** more than 1 TDCP.

[0064] If the number of TRS that the UE **102** receives **312** for the TDCP measurement (e.g., determined based on the TCI state for the PDCCH/PDSCH) is larger than a reported number of TRSs transmitted **306** in the UE capability information, the UE **102** may select one or more TRSs for the TDCP measurement and reporting based on a periodicity of the TRSs, a TCI state ID value, a control resource set (CORESET) ID, or a search space ID.

[0065] The TRS transmitted 312 to the UE 102 for the TDCP measurement may correspond to a configuration for a first TRS associated with 2 CSI-RS resources and/or a second TRS associated with 4 CSI-RS resources. The network entity 304 may configure TRS(s) associated with 2 CSI-RS resources as CMR(s) for TDCP reporting based on periodic TRSs and TRS(s) associated with 4 CSI-RS resources as CMR(s) for TDCP reporting based on aperiodic TRSs. The network entity 304 may refrain from transmitting an IMR configuration for the TDCP measurement. For example, the network entity 304 may refrain from configuring the UE 102 based on csi-IM-ResourcesForInterference or nzp-CSI-RS-ResourcesForInterference in the CSI-report-Config transmitted 308 to the UE 102.

[0066] The network entity 304 may configure TDCP reporting quantities for the TDCP report via RRC parameters (e.g., based on reportQuantity). A multi-path channel between the network entity 304 and the UE 102 may correspond to:

$$h(t) = \sum_{k=1}^K \alpha_k e^{j2\pi f_k t}$$

where α_k corresponds to non-time related parameters, f_k corresponds to the Doppler shift, and K corresponds to a maximum number of paths for the multi-path channel. The multi-path channel can be indicated based on:

$$h(t) = e^{j2\pi(f_{min})t} \sum_{k=1}^K \alpha_k e^{j2\pi(f_k - f_{min})t}$$

$$f_{min} = \min\{f_1, f_2, \dots, f_k\}$$

For CSI reporting, $(f_k - f_{min})$ might correspond to a change in the CSI as a result of the Doppler effect.

[0067] The network entity 304 might configure the UE 102 via the RRC message information element (e.g. reportQuantity) to report the Doppler shift for a strongest path of the multi-path channel based on a channel impulse response (CIR) estimated using the TRS transmitted 312 to the UE 102. In further examples, the network entity might configure the UE 102 via the RRC message parameter (e.g. reportQuantity) to report a maximum Doppler shift of X strongest paths of the multi-path channel based on the CIR estimated using the TRS transmitted 312 to the UE 102, where a value of X may be predefined, configured based on the RRC message, or indicated via MAC-CE or DCI. Alternatively, the UE 102 may determine the value of X based on a received power associated with individual paths of the multi-path channels as well as a threshold power difference from the strongest path. A path that includes a received power above the received power of the strongest path less the threshold power difference may be included in the X strongest paths. The power may be quantified in units of decibel milliwatts (dBm) and threshold may be quantified in the unit of decibel (dB). The threshold may be predefined (e.g., 3 dB) or configured based on RRC messaging via the CSI-reportConfig.

[0068] In further examples, the network entity 304 may configure the UE 102 via the RRC messaging (e.g. reportQuantity) to report the Doppler spread of the X

strongest paths based on the CIR estimated using the TRS transmitted 312 to the UE 102. The delay spread may correspond to a maximum Doppler shift less a minimum Doppler shift of the X strongest paths. In further examples, the network entity 304 may configure the UE 102 via the RRC messaging (e.g. reportQuantity) to report a relative Doppler shift for the strongest path based on the CIR estimated using the TRS transmitted 312 to the UE 102. The relative Doppler shift may correspond to the Doppler shift less the minimum Doppler shift for all the paths of the multi-path channel. In further examples, the network entity 304 may configure the UE 102 via the RRC messaging (e.g. reportQuantity) to report a maximum relative Doppler shift for the X strongest paths based on the CIR estimated using the TRS transmitted 312 to the UE 102. The maximum relative Doppler shift may correspond to the Doppler shift less the minimum Doppler shift for all the paths of the multi-path channel or for the X strongest paths of the multi-path channel.

[0069] The network entity 304 may configure the UE 102 to report both the Doppler shift and the Doppler spread. The network entity 304 may also configure the UE 102 via the RRC messaging (e.g. reportQuantity) to report the Doppler shifts or the relative Doppler shifts for the X strongest paths based on the CIR estimated using the TRS transmitted 312 to the UE 102. The network entity 304 may also configure the UE 102 via the RRC messaging (e.g. reportQuantity) to report a layer 1 (L1)-reference signal receiving power (RSRP) (L1-RSRP) or an L1-SINR that the UE 102 measures 314 based on the TRS transmitted 312 to the UE 102 in addition to the Doppler shift and/or the Doppler spread.

[0070] The network entity 304 and the UE 102 might determine that M CSI processing units are occupied when the UE 102 is configured (e.g., based on the CSI-reportConfig) to measure 314 M TDCPs associated with M TRSs transmitted 312 to the UE 102. A duration of the CSI processing units might be different for different TRSs received 312 from the network entity 304. For example, the CSI processing unit duration for periodic/semi-persistent TDCP reports may begin at a first symbol of the TRS and end at a first symbol of the PUCCH/PUSCH that includes the TRS report or a next symbol after a last symbol of the PUCCH/PUSCH that includes the TRS report. The CSI processing unit duration for aperiodic TDCP reports may begin at a first symbol of the DCI that triggers 310 the TDCP report or a next symbol after a last symbol of the DCI that triggers 310 the TDCP report and end at a first symbol of the PUCCH/PUSCH that includes the TRS report or a next symbol after a last symbol of the PUCCH/PUSCH that includes the TRS report.

[0071] In some examples, the CSI processing units may include a common duration for all M TRSs. The CSI processing unit duration for periodic/semi-persistent TDCP reports may begin at a first symbol for each of the M TRSs and end at a next symbol after a last symbol of the PUCCH/PUSCH that includes the TRS report. The CSI processing unit duration for aperiodic TDCP reports may begin at a first symbol of the DCI that triggers 310 the TDCP report or a next symbol after a last symbol of the DCI that triggers 310 the TDCP report and end at a first symbol of the PUCCH/PUSCH that includes the TDCP report or a next symbol after a last symbol of the PUCCH/PUSCH that includes the TDCP report. In further examples, the network entity 304 and the UE 102 might determine that 1 CSI processing unit

is occupied when the UE 102 is configured (e.g., based on the CSI-reportConfig) to measure 314 the M TDCPs associated with the M TRSs transmitted 312 to the UE 102. For both a periodic/semi-persistent TDCP report and an aperiodic TDCP report, the beginning and ending symbols of the 1 CSI processing unit duration may correspond to the same symbols as used for the CSI processing unit durations associated with the M TRSs.

[0072] The network entity 304 and the UE 102 might determine that X CSI processing units are occupied when the UE 102 is configured (e.g., based on the CSI-reportConfig) to measure 314 the M TDCPs associated with the M TRSs transmitted 312 to the UE 102, where the value of X corresponds to the number of strongest paths in the multipath channel. The UE 102 may determine the value of X based on a number of overlapped TRSs in time domain. Two TRSs may be overlapped in time domain if the TRSs are transmitted 312 at overlapped symbols or overlapped slots. Alternatively, two TRSs may be overlapped in time domain if the TRSs have overlapped time domain durations. A time duration for a TRS may begin at a first symbol of the TRS and end at Y symbols after a last symbol of the TRS, where the UE measures the TDCP within the Y symbols and Y is an integer above 0, which may be predefined or reported by the UE.

[0073] The network entity 304 may also determine the value of X and the value of M based on the UE capability information received 306 from the UE 102. The UE 102 may indicate, in the UE capability information, a UE capability for a maximum number of TRSs per CSI processing unit, which may be denoted by T, such that the network entity 304 may determine the value of X based on $X = \text{ceil}(M/T)$, where "ceil" refers to a function for rounding up to a next largest integer.

[0074] If the network entity 304 triggers 310 multiple CSI-reportConfig associated with more CSI processing units than the maximum number of CSI processing units indicated in the UE capability information, the UE 102 may report non-current CSI or non-current TDCPs for CSI reports with low priority. The priority of the TDCP report transmitted 316 to the network entity 304 may be different than a priority of other types of CSI reports. The network entity 304 and the UE 102 may determine the priority for a CSI report, such as the TDCP report, based on time domain characteristics of the CSI report, such as whether the CSI report is periodic, semi-persistent, or aperiodic. The priority of the CSI report may also be based on the CSI-reportConfigID, the serving cell ID, and/or a reporting quantity (e.g., Doppler shift, Doppler spread, etc.).

[0075] In examples, the priority of the TDCP report transmitted 316 to the network entity 304 can be calculated based on $\text{Pr}_{i_{\text{CSI}}}(y, k, c, s) = 2 \cdot N_{\text{cells}} \cdot M_s \cdot y + N_{\text{cells}} \cdot M_s \cdot k + M_s \cdot c + s$, where an actual priority of the TDCP report is higher if the priority value is lower (e.g., priority 0 corresponds to a higher priority than priority 1). A value of $k=0$ may indicate CSI reports that include L1-RSRP or L1-SINR information, a value of $k=1$ may indicate CSI reports that do not include L1-RSRP, L1-SINR, or TDCP information, a value of $k=2$ may indicate CSI reports that include TDCP information. Further, $y=0$ indicates aperiodic CSI reports transmitted on PUSCH, $y=1$ indicates semi-persistent CSI reports transmitted on the PUSCH, $y=2$ indicates semi-persistent CSI reports transmitted on PUCCH, and $y=3$ indicates periodic CSI reports transmitted on PUCCH. c corresponds to the

serving cell index and N_{cells} corresponds to a value of the higher layer parameter maxNrofServingCells. S corresponds to the reportConfigID and M_s corresponds to a value of the higher layer parameter maxNrofCSI-ReportConfigurations.

[0076] In an alternative example, $k=0$ may indicate CSI reports that include TDCP information, $k=1$ may indicate CSI reports that include L1-RSRP or L1-SINR information, and $k=2$ may indicate CSI reports that do not include L1-RSRP, L1-SINR, or TDCP information. In yet another alternative example, $k=0$ may indicate CSI reports that include L1-RSRP or L1-SINR information, $k=1$ may indicate CSI reports that include TDCP information, and $k=2$ may indicate CSI reports that do not include L1-RSRP, L1-SINR, or TDCP information. In still another alternative example, $k=0$ may indicate CSI reports that include L1-RSRP, L1-SINR, or TDCP information and $k=1$ may indicate CSI reports that do not include L1-RSRP, L1-SINR, or TDCP information.

[0077] The minimum processing delay for the TDCP report may be predefined for the UE 102. For example, the minimum processing delay for the TDCP report can be the same delay as used for an L1-RSRP report (e.g., Z3, Z3'). Alternatively, the minimum processing delay for the TDCP report may be the same delay as used for a CSI report based on a Type1 codebook (e.g., Z1, Z1'). The values of Z1 and Z3 may be based on predefined tables, where Z1 and Z1' indicate a first delay used for a low complexity CSI report and Z3 and Z3' indicate a second delay used for an L1-RSRP report. In further examples, the minimum processing delay may be based on the minimum processing delay for the L1-RSRP report as well as the number of configured M TRSs and the number of CSI processing units for the CSI-reportConfig for a TDCP measurement C. The minimum processing delay may be calculated based on $\text{ceil}(M/C) \cdot Z3$, $\text{ceil}(M/C) \cdot Z3'$. In still further examples, the minimal processing delay for the TDCP report corresponds to the minimum processing delay for a CSI report based on a Type1 codebook as well as the number of configured M TRSs and the number of CSI processing units for the CSI-reportConfig for the TDCP measurement C. Thus, the minimum processing delay may be calculated based on $\text{ceil}(M/C) \cdot Z1$, $\text{ceil}(M/C) \cdot Z1'$.

[0078] The UE 102 may report a UE capability for the minimum processing delay for the TDCP report (Z, Z') in the UE capability information transmitted 306 to the network entity 304. The minimum processing delay for a CSI-reportConfig with M TRSs for a TDCP measurement corresponds to $(M \cdot Z, M \cdot Z')$. In another example, the minimum processing delay for a CSI-reportConfig with M TRSs and C CSI processing units for the TDCP measurement C corresponds to $\text{ceil}(M/C) \cdot Z$, $\text{ceil}(M/C) \cdot Z'$. In a further example associated with a maximum number of configured TRSs, the minimum processing delay for a CSI-reportConfig with M TRSs for the TDCP measurement corresponds to (Z, Z').

[0079] The UE 102 might measure 314 the TDCP based on a wideband operation. For example, the UE 102 might measure 314 the TDCP based on an entire bandwidth for a corresponding TRS. The TDCP, such as the Doppler shift and/or the Doppler spread, may be quantized as a phase shift associated with a certain time duration. In examples, the UE 102 may report multiple bits to indicate the Doppler shift and/or the Doppler spread. Quantization of the multiple bits defines what the multiple bits indicate. For instance, the

Doppler shift and/or the Doppler spread might be quantized based on a phase shift with a range of (0, 360) degrees. Hence, 2 bits might be used to quantize the phase shift, where a step size could be 90 degrees, based on '00': 0-90 degrees, '01': 90-180 degrees; '10': 180-270 degrees; '11': 270-360 degrees. The time duration may be predefined or configured based on the RRC messaging from the network entity 304. A predefined time duration might correspond to a time duration for a slot based on a subcarrier spacing (SCS) for an active uplink/downlink BWP or a subframe. The UE 102 may also determine the time duration based on a symbol interval between two symbols for the TRS received 312 for the TDCP measurement. Alternatively, the TDCP may be quantized based on an absolute value of the Doppler shift and/or the Doppler spread, which may be indicated in hertz (Hz) or kilohertz (kHz). The Doppler shift may be calculated based on $e^{j2\pi f_d t}$ and the Doppler spread may be calculated based on $e^{j2\pi(f_c - f_{min})t}$. The TDCP may be quantized based on a coefficient for the Doppler shift and the Doppler spread calculations as $2\pi f_k$ and $2\pi(f_k - f_{min})$, respectively, or as f_k and $(f_k - f_{min})$, respectively. FIGS. 4A-4C illustrates example TDCP reports that may be generated based on the TDCP measurement and reporting procedure illustrated in FIG. 3.

[0080] FIGS. 4A-4C illustrates tables 410-430 including example TDCP report information transmitted 316 from the UE 102 to the network entity 304. If the UE 102 is configured to transmit 316 the TDCP report (e.g., including the Doppler shift and/or Doppler spread) based on M TRSs for the TDCP measurement, the UE 102 may report V TDCPs in the TDCP report transmitted 316 to the network entity 304, where $0 < V \leq M$. If V is smaller than M, the UE 102 may report V CSI-RS resource set indexes in the TDCP report transmitted 316 to the network entity 304 as well as V quantized TDCPs. A CSI-RS resource set index indicates the resource set from a lowest NZP-CSI-RS-ResourceSetId to a highest NZP-CSI-RS-ResourceSetId of the resource set configured for the TDCP measurement. The TDCP report may be transmitted 316 to the network entity 304 based on an absolute reporting technique associated with a largest or smallest TDCP value, or the TDCP report may be transmitted 316 to the network entity 304 based on a differential reporting technique with a first TDCP being used as a reference point.

[0081] A first table 410 includes example information for a TDCP report that indicates at least one TDCP reporting quantity (e.g., Doppler shift, Doppler spread, etc.) for a TRS. In particular, the first table 410 indicates CSI-RS resource set indexes 1-V when the CSI-RS resource set indexes are reported in the TDCP report transmitted 316 to the network entity 304. The first table 410 also indicates TDCPs for the CSI-RS resource set indexes 1-V when the CSI-RS resource set indexes are reported in the TDCP report transmitted 316 to the network entity 304. Because the TDCP report associated with the first table 410 corresponds to at least one TDCP reporting quantity, at least the TDCP for the CSI-RS resource set index 1 is reported in the TDCP report transmitted 316 to the network entity 304.

[0082] A second table 420 includes example information for a TDCP report that indicates more than one TDCP reporting quantity (e.g., Doppler shift and Doppler spread) for a single TRS. In particular, the second table 420 indicates CSI-RS resource set indexes 1-V when the CSI-RS resource set indexes are reported in the TDCP report transmitted 316

to the network entity 304. The second table 420 also indicates first TDCPs (e.g., Doppler shift) and second TDCPs (e.g., Doppler spread) for the CSI-RS resource set indexes 1-V when the CSI-RS resource set indexes are reported in the TDCP report transmitted 316 to the network entity 304. Because the TDCP report associated with the second table 420 corresponds to more than one TDCP reporting quantity, at least the first TDCP and the second TDCP for the CSI-RS resource set index 1 is recorded in the TDCP report transmitted 316 to the network entity 304.

[0083] A third table 430 includes example information for a TDCP report that indicates a joint TDCP and beam report (e.g., L1-RSRP). In particular, the third table 430 indicates CSI-RS resource set indexes 1-V when the CSI-RS resource set indexes are reported in the TDCP report transmitted 316 to the network entity 304. The third table 430 also indicates TDCPs and L1-RSRP information for the CSI-RS resource set indexes 1-V when the CSI-RS resource set indexes are reported in the TDCP report transmitted 316 to the network entity 304. The L1-RSRPs for the CSI-RS resource set indexes 2-V may correspond to differential L1-RSRP values with respect to the L1-RSRP value for the CSI-RS resource set index 1. Because the TDCP report associated with the third table 430 corresponds to joint TDCP and beam reporting (e.g., L1-RSRP), at least the TDCP and the L1-RSRP for the CSI-RS resource set index 1 is reported in the TDCP report transmitted 316 to the network entity 304. Some TDCP reports may also indicate L1-SINR information.

[0084] In cases where the value of V is the same as the value of M, the UE 102 might not transmit the CSI-RS resource set index to the network entity 304. The value of V may be configured to the UE 102 based on the RRC messaging from the network entity 304, such as by an RRC parameter in the CSI-reportConfig. The UE 102 may determine the value of V based on a number of TCI states associated with the PDCCH/PDSCH. The UE 102 may indicate the value of V in the TDCP report after selecting the TRSs for which to report the beam quality (e.g., L1-RSRP/L1-SINR). The UE 102 measures 314 the selected TRSs and transmits 316 the TDCP report for the TRSs associated with an L1-RSRP/L1-SINR that is higher than a threshold. The threshold may be predefined or configured based on the RRC messaging from the network entity 304. In further examples, the UE 102 may determine the threshold as the L1-RSRP for a strongest TRS minus an offset, where the offset may be predefined or configured based on the RRC messaging from the network entity 304.

[0085] When UE 102 transmits 316 the TDCP report on PUCCH/PUSCH, the value of V may be reported in CSI part 1, and the other information, such as information included in the tables 410-430, may be reported in CSI part 2. If the TDCP report is transmitted 316 with another CSI report, the TDCP may be reported in CSI part 1 or CSI part 2 in different implementations. The CSI part 1 may include a fixed payload size and may be used to identify a number of information bits in the CSI part 2. An entirety of the CSI part 1 is transmitted to the network entity 304 before the CSI part 2 is transmitted to the network entity 304. FIGS. 3-4 illustrate TDCP measurement and reporting procedures as well as example TDCP reports. FIGS. 5-6 show methods for implementing one or more aspects of the TDCP measurement and reporting procedures. In particular, FIG. 5 shows an implementation by the UE 102 of the one or more aspects

of FIGS. 3-4. FIG. 6 shows an implementation by the network entity 304 of the one or more aspects of FIGS. 3-4. [0086] FIG. 5 illustrates a flowchart 500 of a method of wireless communication at a UE. With reference to FIGS. 1 and 7, the method may be performed by the UE 102, the UE apparatus 702, etc., which may include the memory 724' and which may correspond to the entire UE 102 or the UE apparatus 702, or a component of the UE 102 or the UE apparatus 702, such as the wireless baseband processor 724, and/or the application processor 706.

[0087] The UE 102 transmits 506, to a network entity, UE capability information indicative of at least one of: a first UE capability for measurement of a TDCP of a channel between the network entity and the UE or a second UE capability for transmission of a TDCP report to the network entity. For example, referring to FIG. 3, the UE 102 transmits 306, to the network entity 304, the UE capability for TDCP measurement and reporting. The TDCP reporting component 140 of the UE 102 or the UE apparatus 702 may perform the transmission 506.

[0088] The UE 102 receives 508 a TDCP configuration based on the UE capability information—the TDCP configuration indicates a TDCP report structure for at least one of a configured reporting quantity for the TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. For example, referring to FIG. 3, the UE 102 receives 308 a CSI configuration for the TDCP report based on the UE capability transmitted 306, to the network entity 304, for the TDCP measurement and reporting. The TDCP reporting component 140 of the UE 102 or the UE apparatus 702 may perform the reception 508.

[0089] The UE 102 receives 510, from the network entity, a triggering indication for the TDCP report. For example, referring to FIG. 3, the UE 102 receives 310 a TDCP report trigger from the network entity 304. The TDCP reporting component 140 of the UE 102 or the UE apparatus 702 may perform the reception 510.

[0090] The UE 102 receives 512 a TRS from the network entity through the channel between the network entity and the UE. For example, referring to FIG. 3, the UE 102 receives 312 a TRS from the network entity 304. The TDCP reporting component 140 of the UE 102 or the UE apparatus 702 may perform the reception 512.

[0091] The UE 102 measures 514, based on the received TRS, the TDCP of the channel between the network entity and the UE—a measurement of the TDCP of the channel is associated with at least one of: the configured reporting quantity for the TDCP report, the CSI processing unit duration for the TDCP report, or the minimum processing delay for the TDCP report. For example, referring to FIG. 3, based on receiving 312 the TRS from the network entity 304, the UE 102 measures 314 a TDCP of a channel between the network entity 304 and the UE 102 based on a configured TDCP reporting quantity, a CSI processing unit occupancy duration, and/or a minimum processing delay for the TDCP report. The TDCP reporting component 140 of the UE 102 or the UE apparatus 702 may perform the measurement 514.

[0092] The UE 102 transmits 516 the TDCP report to the network entity based on the measurement of the TDCP of the channel. For example, referring to FIG. 3, the UE 102 transmits 316 the TDCP report to the network entity 304 based on the TDCP measurement of the configured TDCP reporting quantity, the CSI processing unit occupancy dura-

tion, and/or the minimum processing delay for the TDCP report. The TDCP reporting component 140 of the UE 102 or the UE apparatus 702 may perform the transmission 516.

[0093] The UE 102 transmits 530, to the network entity, at least one of an L1-RSRP report or an L1-SINR report for the TRS used to measure the TDCP of the channel. For example, referring to FIGS. 3 and 4C, the TDCP report transmitted 316 to the network entity 304 may correspond to the third table 430 that includes L1-RSRP information. The TDCP reporting component 140 of the UE 102 or the UE apparatus 702 may perform the transmission 530. FIG. 5 describes a method from a UE-side of a wireless communication link, whereas FIG. 6 describes a method from a network-side of the wireless communication link.

[0094] FIG. 6 is a flowchart 600 of a method of wireless communication at a network entity. With reference to FIGS. 1 and 8, the method may be performed by the base station 104 or one or more network entities 804 at the base station 104, which may correspond to the RU 106, the DU 108, the CU 110, an RU processor 842, a DU processor 832, a CU processor 812, etc. The base station 104 or the one or more network entities 804 at the base station 104 may include the memory 812'/832'/842', which may correspond to an entirety of the one or more network entities 804 or the base station 104, or a component of the one or more network entities 804 or the base station 104, such as the RU processor 842, the DU processor 832, or the CU processor 812.

[0095] The base station 104 or the one or more network entities at the base station 104 receives 606 UE capability information indicative of at least one of: a first UE capability for a measurement of a TDCP of a channel between a UE and the network entity or a second UE capability for TDCP reporting to a network entity of a TDCP report. For example, referring to FIG. 3, the base station 104 or the one or more network entities at the base station 104 receive 306, from the UE 102, the UE capability for TDCP measurement and reporting. The TDCP report processing component 150 of the base station 104 or the one or more network entities at the base station 104, such as the RU 106, the DU 108, and/or the CU 110 may perform the reception 606.

[0096] The base station 104 or the one or more network entities at the base station 104 transmits 608 a TDCP configuration based on the UE capability information—the TDCP configuration indicates a TDCP report structure for at least one of: a configured reporting quantity for the TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. For example, referring to FIG. 3, the base station 104 or the one or more network entities at the base station 104 transmits 308 a CSI configuration for the TDCP report based on the UE capability received 306, from the UE 102, for the TDCP measurement and reporting. The TDCP report processing component 150 of the base station 104 or the one or more network entities at the base station 104, such as the RU 106, the DU 108, and/or the CU 110 may perform the transmission 608.

[0097] The base station 104 or the one or more network entities at the base station 104 transmits 610, to the UE, a triggering indication for the TDCP report. For example, referring to FIG. 3, the network entity 304 transmits 310 a TDCP report trigger to the UE 102. The TDCP report processing component 150 of the base station 104 or the one

or more network entities at the base station **104**, such as the RU **106**, the DU **108**, and/or the CU **110** may perform the transmission **610**.

[0098] The base station **104** or the one or more network entities at the base station **104** transmits **612** a TRS to the UE through the channel between the UE and the network entity. For example, referring to FIG. 3, the base station **104** or the one or more network entities at the base station **104** transmits **312** a TRS to the UE **102**. The TDCP report processing component **150** of the base station **104** or the one or more network entities at the base station **104**, such as the RU **106**, the DU **108**, and/or the CU **110** may perform the transmission **612**.

[0099] The base station **104** or the one or more network entities at the base station **104** receives **616**, from the UE based on the transmitted TRS, the TDCP report indicative of the measurement of the TDCP of the channel between the UE and the network entity—the measurement of the TDCP of the channel is associated with at least one of: the configured reporting quantity for the TDCP report, the CSI processing unit duration for the TDCP report, or the minimum processing delay for the TDCP report. For example, referring to FIG. 3, the base station **104** or the one or more network entities at the base station **104** receives **316**, from the UE **102** after transmitting **312** the TRS to the UE **102**, the TDCP report associated with the measurement **314** of the configured TDCP reporting quantity, the CSI processing unit occupancy duration, and/or the minimum processing delay for the TDCP report. The TDCP report processing component **150** of the base station **104** or the one or more network entities at the base station **104**, such as the RU **106**, the DU **108**, and/or the CU **110** may perform the reception **616**.

[0100] The base station **104** or the one or more network entities at the base station **104** receives **630**, from the UE, at least one of an L1-RSRP report or an L1-SINR report for the TRS used for the measurement of the TDCP of the channel. For example, referring to FIGS. 3 and 4C, the TDCP report received **316** from the UE **102** may correspond to the third table **430** that includes L1-RSRP information. The TDCP report processing component **150** of the base station **104** or the one or more network entities at the base station **104**, such as the RU **106**, the DU **108**, and/or the CU **110** may perform the reception **630**. A UE apparatus **702**, as described in FIG. 7, may perform the method of flowchart **500**. The base station **104** or the one or more network entities **804** at the base station **104**, as described in FIG. 8, may perform the method of flowchart **600**.

[0101] FIG. 7 is a diagram **700** illustrating an example of a hardware implementation for a UE apparatus **702**. The apparatus **702** may be the UE **102**, a component of the UE **102**, or may implement UE functionality. In some aspects, the apparatus **702** may include a wireless baseband processor **724** (also referred to as a modem) coupled to one or more transceivers **722** (e.g., wireless RF transceiver). The wireless baseband processor **724** may include on-chip memory **724'**. In some aspects, the apparatus **702** may further include one or more subscriber identity modules (SIM) cards **720** and an application processor **706** coupled to a secure digital (SD) card **708** and a screen **710**. The application processor **706** may include on-chip memory **706'**.

[0102] The apparatus **702** may further include a Bluetooth module **712**, a WLAN module **714**, an SPS module **716** (e.g., GNSS module), and a cellular module **717** within the one or more transceivers **722**. The Bluetooth module **712**,

the WLAN module **714**, the SPS module **716**, and the cellular module **717** may include an on-chip transceiver (TRX) (or in some cases, just a receiver (RX)). The Bluetooth module **712**, the WLAN module **714**, the SPS module **716**, and the cellular module **717** may include their own dedicated antennas and/or utilize the antennas **780** for communication. The apparatus **702** may further include one or more sensor modules **718** (e.g., barometric pressure sensor/altimeter; motion sensor such as inertial management unit (IMU), gyroscope, and/or accelerometer(s); light detection and ranging (LIDAR), radio assisted detection and ranging (RADAR), sound navigation and ranging (SONAR), magnetometer, audio and/or other technologies used for positioning), additional modules of memory **726**, a power supply **730**, and/or a camera **732**.

[0103] The wireless baseband processor **724** communicates through the transceiver(s) **722** via one or more antennas **780** with another UE **102s** and/or with an RU associated with a base station **104**. The wireless baseband processor **724** and the application processor **706** may each include a computer-readable medium/memory **724'**, **706'**, respectively. The additional modules of memory **726** may also be considered a computer-readable medium/memory. Each computer-readable medium/memory **724'**, **706'**, **726** may be non-transitory. The wireless baseband processor **724** and the application processor **706** are each responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the wireless baseband processor **724**/application processor **706**, causes the wireless baseband processor **724**/application processor **706** to perform the various functions described. The computer-readable medium/memory may also be used for storing data that is manipulated by the wireless baseband processor **724**/application processor **706** when executing software. The wireless baseband processor **724**/application processor **706** may be a component of the UE **102**. The apparatus **702** may be a processor chip (modem and/or application) and include just the wireless baseband processor **724** and/or the application processor **706**, and in another configuration, the apparatus **702** may be the entire UE **102** and include the additional modules of the apparatus **702**.

[0104] As discussed, the TDCP reporting component **140** is configured to receive a TRS from a network entity through a channel between the network entity and a UE **102**. The TDCP reporting component **140** is further configured to measure, based on the received TRS, a TDCP of the channel between the network entity and the UE **102**. A measurement of the TDCP of the channel is associated with at least one of: a configured reporting quantity for a TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. The TDCP reporting component **140** is further configured to transmit the TDCP report to the network entity based on the measuring the TDCP of the channel. The TDCP reporting component **140** may be within the wireless baseband processor **724**, the application processor **706**, or both the wireless baseband processor **724** and the application processor **706**. The TDCP reporting component **140** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm,

stored within a computer-readable medium for implementation by one or more processors, or some combination thereof.

[0105] As shown, the apparatus **702** may include a variety of components configured for various functions. In one configuration, the apparatus **702**, and in particular the wireless baseband processor **724** and/or the application processor **706**, includes means for receiving a TRS from a network entity through a channel between the network entity and the UE; means for measuring, based on the received TRS, a TDCP of the channel between the network entity and the UE, a measurement of the TDCP of the channel associated with at least one of: a configured reporting quantity for a TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report; and means for transmitting the TDCP report to the network entity based on the measuring. The apparatus **702** further includes means for transmitting, to the network entity, UE capability information indicative of at least one of: a first UE capability for the measuring the TDCP of the channel or a second UE capability for the transmitting the TDCP report to the network entity. The apparatus **702** further includes means for receiving a TDCP configuration based on the UE capability information, the TDCP configuration indicating a TDCP report structure for the at least one of the configured reporting quantity for the TDCP report, the CSI processing unit duration for the TDCP report, or the minimum processing delay for the TDCP report. The apparatus **702** further includes means for transmitting, to the network entity, at least one of an L1-RSRP report or an L1-SINR report for the TRS used to measure the TDCP of the channel. The means for transmitting the TDCP report to the network entity is further configured to: receive, from the network entity, a triggering indication for the TDCP report. The means may be the TDCP reporting component **140** of the apparatus **702** configured to perform the functions recited by the means.

[0106] FIG. **8** is a diagram **800** illustrating an example of a hardware implementation for one or more network entities **804**. The one or more network entities **804** may be a base station, a component of the base station, or may implement base station functionality. The one or more network entities **804** may include at least one of a CU **110**, a DU **108**, or an RU **106**. For example, depending on the layer functionality handled by the TDCP report processing component **150**, the one or more network entities **804** may include the CU **110**; both the CU **110** and the DU **108**; each of the CU **110**, the DU **108**, and the RU **106**; the DU **108**; both the DU **108** and the RU **106**; or the RU **106**.

[0107] The CU **110** may include a CU processor **812**. The CU processor **812** may include on-chip memory **812'**. In some aspects, the CU **110** may further include additional memory modules **814** and a communications interface **818**. The CU **110** communicates with the DU **108** through a midhaul link, such as an F1 interface. The DU **108** may include a DU processor **832**. The DU processor **832** may include on-chip memory **832'**. In some aspects, the DU **108** may further include additional memory modules **834** and a communications interface **838**. The DU **108** communicates with the RU **106** through a fronthaul link. The RU **106** may include an RU processor **842**. The RU processor **842** may include on-chip memory **842'**. In some aspects, the RU **106** may further include additional memory modules **844**, one or

more transceivers **846**, antennas **880**, and a communications interface **848**. The RU **106** communicates wirelessly with the UE **102**.

[0108] The on-chip memory **812'**, **832'**, **842'** and the additional memory modules **814**, **834**, **844** may each be considered a computer-readable medium/memory. Each computer-readable medium/memory may be non-transitory. Each of the processors **812**, **832**, **842** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the corresponding processor(s) causes the processor(s) to perform the various described functions. The computer-readable medium/memory may also be used for storing data that is manipulated by the processor(s) when executing software.

[0109] As discussed, the TDCP report processing component **150** is configured to transmit a TRS to a UE through a channel between the UE and the network entity. The TDCP report processing component **150** is further configured to receive, from the UE based on the transmitted TRS, a TDCP report indicative of a measurement of the TDCP of the channel between the UE and the network entity. The measurement of the TDCP of the channel is associated with at least one of: a configured reporting quantity for the TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. The TDCP report processing component **150** may be within one or more processors of one or more of the CU **110**, DU **108**, and the RU **106**. The TDCP report processing component **150** may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by one or more processors configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by one or more processors, or some combination thereof.

[0110] The one or more network entities **804** may include a variety of components configured for various functions. In one configuration, the one or more network entities **804** include means for transmitting a TRS to a UE through a channel between the UE and the network entity; and means for receiving, from the UE based on the transmitted TRS, a TDCP report indicative of a measurement of the TDCP of the channel between the UE and the network entity, the measurement of the TDCP of the channel associated with at least one of: a configured reporting quantity for the TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report. The one or more network entities **804** further include means for receiving UE capability information indicative of at least one of: a first UE capability for the measurement of the TDCP of the channel or a second UE capability for TDCP reporting of the TDCP report to the network entity. The one or more network entities **804** further include means for transmitting a TDCP configuration based on the UE capability information, the TDCP configuration indicating a TDCP report structure for the at least one of: the configured reporting quantity for the TDCP report, the CSI processing unit duration for the TDCP report, or the minimum processing delay for the TDCP report. The one or more network entities **804** further include means for receiving, from the UE, at least one of an L1-RSRP report or an L1-SINR report for the TRS used for the measurement of the TDCP of the channel. The means for receiving the TDCP report from the UE is further configured to: transmit, to the UE, a triggering

indication for the TDCP report. The means may be the TDCP report processing component **150** of the one or more network entities **804** configured to perform the functions recited by the means.

[0111] The specific order or hierarchy of blocks in the processes and flowcharts disclosed herein is an illustration of example approaches. Hence, the specific order or hierarchy of blocks in the processes and flowcharts may be rearranged. Some blocks may also be combined or deleted. Optional blocks of the processes and flowcharts may be indicated by dashed lines. The accompanying method claims present elements of the various blocks in an example order, and are not limited to the specific order or hierarchy presented in the claims, processes, and flowcharts.

[0112] The detailed description set forth herein describes various configurations in connection with the drawings and does not represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough explanation of various concepts. However, these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

[0113] Aspects of wireless communication systems, such as telecommunication systems, are presented with reference to various apparatuses and methods. These apparatuses and methods are described in the following detailed description and are illustrated in the accompanying drawings by various blocks, components, circuits, processes, call flows, systems, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or combinations thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

[0114] An element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems-on-chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other similar hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software, which may be referred to as software, firmware, middleware, microcode, hardware description language, or otherwise. Software shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, or any combination thereof.

[0115] If the functionality described herein is implemented in software, the functions may be stored on, or encoded as, one or more instructions or code on a computer-readable medium, such as a non-transitory computer-readable storage medium. Computer-readable media includes computer storage media and can include a random-access

memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of these types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer. Storage media may be any available media that can be accessed by a computer.

[0116] Aspects, implementations, and/or use cases described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, the aspects, implementations, and/or use cases may come about via integrated chip implementations and other non-module-component based devices, such as end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, machine learning (ML)-enabled devices, etc. The aspects, implementations, and/or use cases may range from chip-level or modular components to non-modular or non-chip-level implementations, and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more techniques described herein.

[0117] Devices incorporating the aspects and features described herein may also include additional components and features for the implementation and practice of the claimed and described aspects and features. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes, such as hardware components, antennas, RF-chains, power amplifiers, modulators, buffers, processor(s), interleavers, adders/summers, etc. Techniques described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc., of varying configurations.

[0118] The description herein is provided to enable a person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not limited to the aspects described herein, but are to be interpreted in view of the full scope of the present disclosure consistent with the language of the claims.

[0119] Reference to an element in the singular does not mean “one and only one” unless specifically stated, but rather “one or more.” Terms such as “if,” “when,” and “while” do not imply an immediate temporal relationship or reaction. That is, these phrases, e.g., “when,” do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. Unless specifically stated otherwise, the term “some” refers to one or more. Combinations such as “at least one of A, B, or C” or “one or more of A, B, or C” include any combination of A, B, and/or C, such as A and B, A and C, B and C, or A and B and C, and may include multiples of A, multiples of B, and/or multiples of C, or may include A only, B only, or C only. Sets should be interpreted as a set of elements where the elements number one or more.

[0120] Structural and functional equivalents to elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are encompassed by the claims. The words “module,” “mechanism,” “element,” “device,” and the like may not be a substitute for the word “means.” As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.” As used herein, the phrase “based on” shall not be construed as a reference to a closed set of information, one or more conditions, one or more factors, or the like. In other words, the phrase “based on A”, where “A” may be information, a condition, a factor, or the like, shall be construed as “based at least on A” unless specifically recited differently.

[0121] The following examples are illustrative only and may be combined with other examples or teachings described herein, without limitation.

[0122] Example 1 is a method of wireless communication at a UE, including: receiving a TRS from a network entity through a channel between the network entity and the UE; measuring, based on the received TRS, a TDCP of the channel between the network entity and the UE, a measurement of the TDCP of the channel associated with at least one of: a configured reporting quantity for a TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report; and transmitting the TDCP report to the network entity based on the measuring of the TDCP of the channel.

[0123] Example 2 may be combined with example 1 and further includes transmitting, to the network entity, UE capability information indicative of at least one of: a first UE capability for the measuring the TDCP of the channel or a second UE capability for the transmitting the TDCP report to the network entity.

[0124] Example 3 may be combined with any of examples 1-2 and further includes receiving a TDCP configuration based on the UE capability information, the TDCP configuration indicating a TDCP report structure for the at least one of the configured reporting quantity for the TDCP report, the CSI processing unit duration for the TDCP report, or the minimum processing delay for the TDCP report.

[0125] Example 4 may be combined with example 3 and includes that the TDCP configuration is received as part of an RRC message and corresponds to a set of available TRSs for the measuring the TDCP of the channel, and includes that at least a subset of TRSs of the set of available TRSs is selected for the measuring the TDCP of the channel based on at least one of a MAC-CE or DCI.

[0126] Example 5 may be combined with any of examples 1-4 and includes that the configured reporting quantity for the TDCP report corresponds to at least one of: a first maximum number of TRSs per TDCP configuration, a second maximum number of TRSs across a plurality of TDCP configurations, or a third maximum number of TDCP configurations for the TDCP report.

[0127] Example 6 may be combined with any of examples 1-5 and includes that the TRS is configured in a TCI state for at least one of a PDCCH or a PDSCH associated with the TRS.

[0128] Example 7 may be combined with any of examples 1-6 and includes that the TDCP report is indicative of at least one of: a Doppler shift for one or more strongest paths, a

maximum Doppler shift for a strongest path of the one or more strongest paths, a relative Doppler shift for the one or more strongest paths, a maximum relative Doppler shift for the strongest path of the one or more strongest paths, or a Doppler spread for the one or more strongest paths.

[0129] Example 8 may be combined with any of examples 1-7 and includes that the one or more strongest paths are based on a CIR associated with the TRS.

[0130] Example 9 may be combined with example 7 and includes that a number of the one or more strongest paths is at least one of: configured based on the RRC message, configured based on the MAC-CE, indicated based on the DCI, based on a predefined protocol, based on an RSRP associated with a plurality of paths that include the one or more strongest paths, or based on a threshold power difference from the strongest path of the one or more strongest paths.

[0131] Example 10 may be combined with any of examples 1-9 and further includes transmitting, to the network entity, at least one of an L1-RSRP report or an L1-SINR report for the TRS used to measure the TDCP of the channel.

[0132] Example 11 may be combined with any of examples 1-10 and includes that a first number of CSI processing units occupied by the UE during the CSI processing unit duration is based on a second number of TDCPs of the channel measured by the UE for a corresponding number of TRSs.

[0133] Example 12 may be combined with any of examples 1-11 and includes that the transmitting the TDCP report to the network entity further includes: receiving, from the network entity, a triggering indication for the TDCP report.

[0134] Example 13 may be combined with any of examples 1-12 and includes that the TDCP report is prioritized relative to at least one of the L1-RSRP report, the L1-SINR report, or a CSI report that excludes the L1-RSRP report, the L1-SINR report, and the TDCP report.

[0135] Example 14 may be combined with any of examples 1-13 and includes that the minimum processing delay for the TDCP report is based on a processing time for at least one of the CSI report, the L1-RSRP report, the first number of CSI processing units, or the corresponding number of TRSs.

[0136] Example 15 may be combined with any of examples 4-13 and includes that the TDCP report indicates a first total number of TDCPs of the channel that is less than or equal to a second total number of the at least the subset of TRSs of the set of available TRSs selected for the measuring the TDCP of the channel.

[0137] Example 16 may be combined with examples 15 and includes that the TDCP report further indicates at least one of: a corresponding first total number of CSI-RS resource set indexes, a beam report, or the CSI report.

[0138] Example 17 may be combined with any of examples 1-16 and includes that the TDCP report is associated with at least one of a phase shift over a time duration, a first absolute value of the Doppler shift, a second absolute value of the Doppler spread, or a coefficient for calculation of the phase shift over the time duration.

[0139] Examples 18 is a method of wireless communication at a network entity, including: transmitting a TRS to a UE through a channel between the UE and the network entity; and receiving, from the UE based on the transmitted

TRS, a TDCP report indicative of a measurement of the TDCP of the channel between the UE and the network entity, the measurement of the TDCP of the channel associated with at least one of: a configured reporting quantity for the TDCP report, a CSI processing unit duration for the TDCP report, or a minimum processing delay for the TDCP report.

[0140] Example 19 may be combined with example 18 and further includes receiving UE capability information indicative of at least one of: a first UE capability for the measurement of the TDCP of the channel or a second UE capability for TDCP reporting of the TDCP report to the network entity.

[0141] Example 20 may be combined with any of examples 18-19 and further includes transmitting a TDCP configuration based on the UE capability information, the TDCP configuration indicating a TDCP report structure for the at least one of: the configured reporting quantity for the TDCP report, the CSI processing unit duration for the TDCP report, or the minimum processing delay for the TDCP report.

[0142] Example 21 may be combined with example 20 and includes that the TDCP configuration is transmitted as part of an RRC message and corresponds to a set of available TRSs for the measurement of the TDCP of the channel, and includes that at least a subset of TRSs of the set of available TRSs is selected for the measurement of the TDCP of the channel based on at least one of a MAC-CE or DCI.

[0143] Example 22 may be combined with any of examples 18-21 and includes that the configured reporting quantity for the TDCP report corresponds to at least one of: a first maximum number of TRSs per TDCP configuration, a second maximum number of TRSs across a plurality of TDCP configurations, or a third maximum number of TDCP configurations for the TDCP report.

[0144] Example 23 may be combined with any of examples 18-22 and includes that the TRS is configured in a TCI state for at least one of a PDCCH or a PDSCH associated with the TRS.

[0145] Example 24 may be combined with any of examples 18-23 and includes that the TDCP report is indicative of at least one of: a Doppler shift for one or more strongest paths, a maximum Doppler shift for a strongest path of the one or more strongest paths, a relative Doppler shift for the one or more strongest paths, a maximum relative Doppler shift for the strongest path of the one or more strongest paths, or a Doppler spread for the one or more strongest paths.

[0146] Example 25 may be combined with any of examples 18-24 and includes that the one or more strongest paths are based on a CIR associated with the TRS.

[0147] Example 26 may be combined with example 24 and includes that a number of the one or more strongest paths is at least one of: configured based on the RRC message, configured based on the MAC-CE, indicated based on the DCI, based on a predefined protocol, based on an RSRP associated with a plurality of paths that include the one or more strongest paths, or based on a threshold power difference from the strongest path of the one or more strongest paths.

[0148] Example 27 may be combined with any of examples 18-26 and further includes receiving, from the UE, at least one of an L1-RSRP report or an L1-SINR report for the TRS used for the measurement of the TDCP of the channel.

[0149] Example 28 may be combined with any of examples 18-27 and includes that a first number of CSI processing units occupied during the CSI processing unit duration is based on a second number of TDCPs of the channel measured for a corresponding number of TRSs.

[0150] Example 29 may be combined with any of examples 18-28 and includes that the receiving the TDCP report from the UE further includes: transmitting, to the UE, a triggering indication for the TDCP report.

[0151] Example 30 may be combined with any of examples 18-29 and includes that the TDCP report is prioritized relative to at least one of the L1-RSRP report, the L1-SINR report, or a CSI report that excludes the L1-RSRP report, the L1-SINR report, and the TDCP report.

[0152] Example 31 may be combined with any of examples 18-30 and includes that the minimum processing delay for the TDCP report is based on a processing time for at least one of the CSI report, the L1-RSRP report, the first number of CSI processing units, or the corresponding number of TRSs.

[0153] Example 32 may be combined with any of examples 18-31 and includes that the TDCP report indicates a first total number of TDCPs of the channel that is less than or equal to a second total number of the at least the subset of TRSs of the set of available TRSs selected for the measurement of the TDCP of the channel.

[0154] Example 33 may be combined with example 32 and includes that the TDCP report further indicates at least one of: a corresponding first total number of CSI-RS resource set indexes, a beam report, or the CSI report.

[0155] Example 34 may be combined with any of examples 18-33 and includes that the TDCP report is associated with at least one of a phase shift over a time duration, a first absolute value of the Doppler shift, a second absolute value of the Doppler spread, or a coefficient for calculation of the phase shift over the time duration.

[0156] Example 35 is an apparatus for wireless communication for implementing a method as in any of examples 1-34.

[0157] Example 36 is an apparatus for wireless communication including means for implementing a method as in any of examples 1-34.

[0158] Example 37 is a non-transitory computer-readable medium storing computer executable code, the code when executed by at least one processor causes the at least one processor to implement a method as in any of examples 1-34.

What is claimed is:

1. A method of wireless communication at a user equipment (UE), comprising:

receiving, from a network entity, a channel state information (CSI) report configuration configuring a reporting quantity set to time domain channel property (TDCP) for a TDCP report, the CSI report configuration indicating a tracking reference signal (TRS) as a channel measurement resource (CMR) and omitting an interference measurement resource (IMR);

receiving, from the network entity, the TRS based on the CSI report configuration;

measuring, based on the received TRS, the TDCP of a channel between the network entity and the UE; and transmitting, to the network entity, the TDCP report based on the measuring of the TDCP of the channel.

2. The method of claim 1, further comprising transmitting, to the network entity, UE capability information indicative of at least one of: a first UE capability for the measuring the TDCP of the channel or a second UE capability for the transmitting the TDCP report to the network entity.

3. The method of claim 2, wherein the CSI report configuration is based on the UE capability information, the CSI report configuration indicating a TDCP report structure for at least one of the configured reporting quantity for the TDCP report or a CSI processing unit duration for the TDCP report, wherein the measuring of the TDCP of the channel is associated with the CSI processing unit duration for the TDCP report.

4. The method of claim 3, wherein the CSI report configuration is received as part of a radio resource control (RRC) message and corresponds to a set of available TRSs for the measuring the TDCP of the channel, and wherein at least a subset of TRSs of the set of available TRSs is selected for the measuring the TDCP of the channel based on at least one of a medium access control-control element (MAC-CE) or downlink control information (DCI).

5. The method of claim 1, wherein the TRS corresponds to a channel state information-reference signal (CSI-RS) resource set, the method further comprising:

transmitting, to the network entity, UE capability information indicating a first maximum number of CSI-RS resources per CSI report configuration for the TDCP report and a second maximum number of CSI-RS resources across a plurality of CSI report configurations for the TDCP report.

6. The method of claim 5, wherein the first maximum number of the CSI-RS resources and the second maximum number of the CSI-RS resources are configured per component carrier (CC) or per frequency band.

7. The method of claim 1, wherein the TRS is configured in a transmission configuration indicator, TCI, state for at least one of a physical downlink control channel (PDCCH) or a physical downlink shared channel (PDSCH) associated with the TRS.

8. The method of claim 1, wherein the TDCP report is indicative of at least one of: a Doppler shift for one or more strongest paths, a maximum Doppler shift for a strongest path of the one or more strongest paths, a relative Doppler shift for the one or more strongest paths, a maximum relative Doppler shift for the strongest path of the one or more strongest paths, or a Doppler spread for the one or more strongest paths.

9. (canceled)

10. (canceled)

11. The method of claim 1, further comprising transmitting, to the network entity, at least one of a layer 1 RSRP (L1-RSRP), report or an L1 signal-to-interference-plus-noise ratio (L1-SINR) report for the TRS used to measure the TDCP of the channel.

12. The method of claim 1, wherein a first number of CSI processing units occupied by the UE during the CSI processing unit duration is based on a second number of TDCPs of the channel measured by the UE for a corresponding number of TRSs and a UE capability.

13. The method of claim 1, wherein the transmitting the TDCP report to the network entity further comprises:

receiving, from the network entity, a triggering indication for the TDCP report.

14. The method of claim 1, wherein the TDCP report is prioritized relative to at least one of:

the L1-RSRP report,

the L1-SINR report, or

a CSI report that excludes the L1-RSRP report, the L1-SINR report, and the TDCP report.

15. The method of claim 4, wherein the TDCP report indicates a first total number of TDCPs of the channel that is less than or equal to a second total number of the TRSs selected for the measuring the TDCP of the channel.

16. and wherein the TDCP report further indicates at least one of: a corresponding first total number of CSI-reference signal (CSI-RS) resource set indexes, a beam report, or the CSI report.

17. The method of claim 1, wherein the TDCP report is associated with at least one of a phase shift over a time duration, a first absolute value of the Doppler shift, a second absolute value of the Doppler spread, or a coefficient for calculation of the phase shift over the time duration.

18. The method of claim 1, wherein the measuring comprises measuring the TDCP based on a wideband operation.

19. A method of wireless communication at a network entity, comprising:

transmitting, to a user equipment (UE), a channel state information (CSI) report configuration configuring a reporting quantity set to time domain channel property (TDCP) for a TDCP report, the CSI report configuration indicating a tracking reference signal (TRS) as a channel measurement resource (CMR) and omitting an interference measurement resource (IMR);

transmitting, to the UE, the TRS based on the CSI report configuration; and

receiving, from the UE, the TDCP report indicating a measurement of the TDCP of a channel between the UE and the network entity, the measurement of the TDCP of the channel being based on the TRS.

20. The method of claim 16, further comprising receiving (306, 606) UE capability information indicating at least one of: a first UE capability for the measurement of the TDCP of the channel or a second UE capability for TDCP reporting of the TDCP report to the network entity.

21. (canceled)

22. The method of claim 16, wherein the TRS corresponds to a channel state information-reference signal (CSI-RS) resource set, the method further comprising:

receiving, from the UE, UE capability information indicating a first maximum number of CSI-RS resources per CSI report configuration for the TDCP report and a second maximum number of CSI-RS resources across a plurality of CSI report configurations for the TDCP report, wherein the first maximum number of the CSI-RS resources and the second maximum number of the CSI-RS resources are configured per component carrier (CC) or per frequency band.

23. An apparatus for wireless communication, comprising:

a processor; and

a memory coupled to the processor to store instructions, which when executed by the processor, cause the apparatus to:

receive, from a network entity, a channel state information (CSI) report configuration configuring a reporting quantity set to time domain channel property (TDCP) for a TDCP report, the CSI report

configuration indicating a tracking reference signal (TRS) as a channel measurement resource (CMR) and omitting an interference measurement resource (IMR);

receive, from the network entity, the TRS based on the CSI report configuration;

measure, based on the received TRS, the TDCP of a channel between the network entity and the UE; and

transmit, to the network entity, the TDCP report based on the measuring of the TDCP of the channel.

24. The method of claim **169**, wherein the TRS corresponds to a channel state information-reference signal (CSI-RS) resource set, wherein the instructions, which when executed by the processor, further causes the apparatus to:

transmit, from the UE, UE capability information indicating a first maximum number of CSI-RS resources per CSI report configuration for the TDCP report and a second maximum number of CSI-RS resources across a plurality of CSI report configurations for the TDCP report, wherein the first maximum number of the CSI-RS resources and the second maximum number of the CSI-RS resources are configured per component carrier (CC) or per frequency band.

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